

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Permits Section, WRD, EGLE
Date:	8/1/2023

Outf.	Authorized Flow		Disch. Type *	Receiving Water	Lake?	FLOWS (cfs)			pH	Hardness
	mgd	cfs				95% Exc.	H. Mean	90Q10		
001	3.2	4.95		Clam River	☐	2.6	13.0	3.4	7	70

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

TYPE OF REVIEW: NPDES Permit Reissuance

LOW FLOW FILE NO.: 10194 (Wexford County)

HARDNESS SOURCE: STORET 830054 collected on August 19, 2021 by Permits Section staff.

PEL SHEET VERSION: 9-26-2022

Review the Following Information Sources

☐ Permit Application

☐ Facility/VGW File

☐ MiWaters Documents & Files

☐ Current Permit

☐ WTA's

☐ Staff Report #'s

☐ WET Tests

☐ Other

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	70 mg/L CaCO3
Outfall:	001	pH:	7
Date:	8/1/2023	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met. Translator	Background Conc.
		FCV	vd	HNH	vd	HCV	vd	WV	vd	FAV	vd		
Perfluorooctane sulfonate @	1763231	140	2201408	0.012	1201403	NA		NA		1600	2201408	1	
Perfluorooctanoic acid	335671	880	2201007	0.17	1202012	NA		NA		15000	2201007	1	
Perfluorobutanesulfonate	375735	24000	1201112	670	1202112	NA		NA		240000	1201112	1	
Perfluorooctane sulfonate @	1763231	140	2201408	0.012	1201403	NA		NA		1600	2201408	1	
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.6 cfs	Conc.	Load (lbs/day)
Outfall:	001			H. Mean Flow:	13 cfs	Hg Loading Calculated based on LCA (ng/L) of :	0
Date:	8/1/2023			90Q10 Flow:	3.4 cfs	TP Loading based on concentration (mg/L) of :	0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Perfluorooctane sulfonate @	140	3.73632	0.012	0.0003203	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1600	42.7008	0.01326	0.01326	Reasonable Potential. Review data using Heaton May 29, 2020 Guidance. See review below:	
Perfluorooctanoic acid	995.4725	26.56717	0.281535938	0.0075136	#VALUE!	#VALUE!	#VALUE!	#VALUE!	15000	400.32	0.036	0.036		
Perfluorobutanesulfonate	27149.25	724.55918	1109.582813	29.612546	#VALUE!	#VALUE!	#VALUE!	#VALUE!	240000	6405.12	0.0078003	0.0169072	No Reasonable Potential. No Recommendation.	
											0.1044	0.1044		
Perfluorooctane sulfonate @	140	3.73632	0.012	0.0003203	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1600	42.7008	0.0323904	0.0584242	No Reasonable Potential. No Recommendation.	
											0.01404	0.01404		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0.0037799	0.0072464		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

* SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY

Parameter:	Perfluorooctane sulfonate @	Source	Perfluorooctanoic acid	Source	Perfluorobutanesulfonate	Source	Perfluorooctane sulfonate @	Source
No. Nondetect	2		0		0		0	
Data	0.0065 11/5/2018 0.0078 6/4/2019 0.002 10/22/2019 0.0027 4/22/2020 0.0018 10/7/2020 0.0031 4/21/2022 0.0014 10/13/2022 0.00188 4/20/2023		0.02 11/5/2018 0.016 6/4/2019 0.0034 10/22/2019 0.0047 4/22/2020 0.0025 10/7/2020 0.0019 4/13/2021 0.002 10/13/2021 0.002 4/21/2022 0.0016 10/13/2022 0.002 4/20/2023		0.058 11/5/2018 0.049 6/4/2019 0.018 10/22/2019 0.0031 4/22/2020 0.025 10/7/2020 0.019 4/13/2021 0.019 10/13/2021 0.032 4/21/2022 0.034 10/13/2022 0.0145 4/20/2023		0.0065 11/5/2018 0.0078 6/4/2019 0.002 10/22/2019 0.0027 4/22/2020 0.0018 10/7/2020 0.0031 4/21/2022 0.0014 10/13/2022 0.00188 4/20/2023 0.0016 QL data from below 0.0019 QL data from below	
	Not Detected 4/13/2021 (<1.6 ng/l) 10/13/2021 (<1.9 ng/l)						Not Detected 4/13/2021 (<1.6 ng/l) 10/13/2021 (<1.9 ng/l)	
							(QL <#. ug/l)	
							The QL for the Not Detected data was substituted as a quantified result based on guidance from the May 29, 2020 Heaton Memo titled: "Water Resources Division (WRD) Guidance for Perfluorooctanesulfonic acid (PFOS) and Perfluorooctanoic acid (PFOA) Reasonable Potential Evaluations".	
							Option 4. was used for the analysis above.	

PEQ -- Effluent Data Summary
(all values in ug/L)

[illegible]

C

EGLE
Page 6
MI0020257 v7.0
Approved
Issued On:11/29/2023
Expires On:10/01/2027

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Perfluorooctane sulfonate @	Perfluorooctanoic acid	Perfluorobutanesulfonate	Perfluorooctane sulfonate @	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
Sample Max.	0.0078	0.02	0.058	0.0078	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Samples	10	9	9	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Multiplier	1.7	1.8	1.8	1.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PEQ:	0.01326	0.036	0.1044	0.01404	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PROJECTED EFFLUENT QUALITY ANALYSIS (>=10 Detectable):

PARAMETER:	Perfluorooctane sulfonate @		Perfluorooctanoic acid		Perfluorobutanesulfonate		Perfluorooctane sulfonate @
TOTAL NO. VALUES:	10		10		10		10
DETECTED:	8		10		10		10
NON-DETECTED:	2		0		0		0
d (% data<detect)	0.2		0		0		0
m (mean of data >detect)	0.0033975		0.00561		0.02716		0.003068
std dev	0.0024017955		0.0066612395		0.0165301207		0.0022303004
n	1	30	1	30	1	30	1
d^n	0.2	1.07E-021	0	0	0	0	0
p	0.9375	0.95	0.95	0.95	0.95	0.95	0.95
std normal dist fn (Z)	2.354820045	2.4477468	2.4477468307	2.4477468	2.4477468307	2.4477468	2.4477468307
Z_p	1.5344347326	1.6452114	1.6452114401	1.6452114	1.6452114401	1.6452114	1.6452114401
1+(s/m)^2	1.4997503513	1.4997504	2.4098872052	2.4098872	1.3704183696	1.3704184	1.5284647512
(sigma_d)^2	0.4052986618	0.4052987	0.8795799436	0.8795799	0.3151160725	0.3151161	0.424263801
mu_d	-5.8873647428	-5.887365	-5.6229945312	-5.622995	-3.7635680126	-3.763568	-5.998861296
(sigma_dn)^2	0.4052986618	0.0287393	0.8795799436	0.0459253	0.3151160725	0.0122717	0.424263801
mu_dn	-5.8873647428	-5.922229	-5.6229945312	-5.206167	-3.7635680126	-3.612146	-5.998861296
P_95 exponent	-4.9104964685	-5.643321	-4.0800177895	-4.853595	-2.8400252734	-3.429893	-4.9272440111
P_95 (PEQ)	0.007368829	0.0035411	0.0169071649	0.0078003	0.0584241894	0.0323904	0.0072464469
	Max	Average	Max	Average	Max	Average	Max

NOTES: For purposes of this summary, ^ represents an exponent or superscript while _ represents a subscript.
Variables are defined in Part 12, Rule 1225 (3) (a).



	#N/A		#N/A		#N/A		#N/A		#N/A		#N/A		#N/A	
	0		0		0		0		0		0		0	
	0		0		0		0		0		0		0	
	0		0		0		0		0		0		0	
	#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
	#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
	#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
30	1	30	1	30	1	30	1	30	1	30	1	30	1	30
0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.95	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
2.4477468	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
1.6452114	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
1.5284648	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.4242638	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
-5.998861	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.0174621	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
-5.79546	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
-5.578055	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.0037799	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

CAS #	PARAMETER NAME
4860031	1-Chlorohexadecane
90120	1-Methylnaphthalene #
75343	1,1-Dichloroethane
75354	1,1-Dichloroethylene
71556	1,1,1-Trichloroethane
630206	1,1,1,2-Tetrachloroethane #
79005	1,1,2-Trichloroethane #
76131	1,1,2-Trichloro-1,2,2-trifluoroethane
79345	1,1,2,2-Tetrachloroethane #
632224	1,1,3,3-Tetramethylurea
112265	1,2-Bis(2-chloroethoxy)ethane
96128	1,2-Dibromo-3-chloropropane #
106934	1,2-Dibromoethane #
95501	1,2-Dichlorobenzene
107062	1,2-Dichloroethane #
540590	1,2-Dichloroethylene
156592	<i>cis</i> -1,2-Dichloroethylene
156605	<i>trans</i> -1,2-Dichloroethylene
78875	1,2-Dichloropropane #
542756	1,3-Dichloropropylene #
10061015	<i>cis</i> -1,3-Dichloropropylene #
10061026	<i>trans</i> -1,3-Dichloropropylene #
87616	1,2,3-Trichlorobenzene
526738	1,2,3-Trimethylbenzene
120821	1,2,4-Trichlorobenzene
95636	1,2,4-Trimethylbenzene
634662	1,2,3,4-Tetrachlorobenzene @
95943	1,2,4,5-Tetrachlorobenzene @
541731	1,3-Dichlorobenzene
108678	1,3,5-Trimethylbenzene
106467	1,4-Dichlorobenzene #
123911	1,4-Dioxane #
575371	1,7-Dimethylnaphthalene
95512	2-Chloroaniline
95578	2-Chlorophenol
104767	2-Ethylhexanol #
611143	2-Ethyltoluene
591786	2-Hexanone
88697	2-Isopropylphenol
91576	2-Methylnaphthalene
95487	2-Methylphenol
88755	2-Nitrophenol
67630	2-Propanol
594207	2,2-Dichloropropane
77769	2,2-Dimethoxypropane
526750	2,3-Dimethylphenol
58902	2,3,4,6-Tetrachlorophenol
1746016	2,3,7,8-TCDD # @
94757	2,4-D

120832	2,4-Dichlorophenol
105679	2,4-Dimethylphenol
51285	2,4-Dinitrophenol
120956	2,4-Di- <i>tert</i> -pentylphenol
93765	2,4,5-T
88062	2,4,6-Trichlorophenol #
126863	2,4,7,9-Tetramethyl-5-decyne-4,7-diol
585342	3- <i>tert</i> -Butylphenol
620144	3-Ethyltoluene
618451	3-Isopropylphenol
108394	3-Methylphenol
91941	3,3'-Dichlorobenzidine #
95658	3,4-Dimethylphenol
40360449	3,5,6-Trichloropicolinic acid
1702176	3,6-Dichloropicolinic acid
59507	4-Chloro-3-methylphenol
106478	4-Chloroaniline #
98668	4-Chlorobenzene sulfonic acid
106489	4-Chlorophenol
99898	4-Isopropylphenol
108112	4-Methyl-2-pentanol
108101	4-Methyl-2-pentanone
128370	4-Methyl-2,6-di-T-butylphenol #
106445	4-Methylphenol
100027	4-Nitrophenol
84852153	4-Nonylphenol
4684940	6-Chloropicolinic acid
208968	Acenaphthylene
83329	Acenaphthene
75070	Acetaldehyde
71501	Acetate (includes acetic acid)
64197	Acetic acid (includes acetate)
67641	Acetone
75058	Acetonitrile
98862	Acetophenone
79061	Acrylamide #
107131	Acrylonitrile #
15972608	Alachlor #
309002	Aldrin #
7664417	Ammonia, total ammonia nitrogen
62533	Aniline #
120127	Anthracene
7440360	Antimony
7440382	Arsenic #
1912249	Atrazine
103333	Azobenzene #
495487	Azoxybenzene
7440393	Barium
71432	Benzene #
92875	Benzidine #

56553	Benzo(a)anthracene
50328	Benzo(a)pyrene #
205992	Benzo(b)fluoranthene #
191242	Benzo(g,h,i)perylene
7440417	Beryllium
92524	Biphenyl
80057	Bisphenol A
111444	Bis(2-chloroethyl)ether #
108601	Bis(2-chloroisopropyl)ether #
117817	Bis(2-ethylhexyl)phthalate #
7440428	Boron
15541454	Bromate #
7726956	Bromine
74975	Bromochloromethane
75274	Bromodichloromethane #
75252	Bromoform #
74839	Bromomethane
71363	<i>n</i> -Butanol
85687	Butyl benzyl phthalate
109739	Butylamine
104518	<i>n</i> -Butylbenzene
135988	<i>sec</i> -Butylbenzene
98066	<i>tert</i> -Butylbenzene
7440439	Cadmium
63252	Carbaryl
86748	Carbazole #
75150	Carbon disulfide
56235	Carbon tetrachloride #
57749	Chlordane # @
16887006	Chloride
7782505	Chlorine
14998277	Chlorite
108907	Chlorobenzene
75003	Chloroethane #
67663	Chloroform #
74873	Chloromethane #
95498	<i>o</i> -Chlorotoluene
2921882	Chlorpyrifos
7440473	Chromium
18540299	Chromium, hexavalent
218019	Chrysene #
7440484	Cobalt
7440508	Copper
21725462	Cyanazine #
57125	Cyanide, free
110827	Cyclohexane
221667318	Cyprosulfamide
10222012	DBNPA
50293	DDT # @
333415	Diazinon

53703	Dibenz(a,h)anthracene #
132649	Dibenzofuran
124481	Dibromochloromethane #
111922	Dibutylamine
75718	Dichlorodifluoromethane
75274	Dichlorobromomethane #
75434	Dichlorofluoromethane
62737	Dichlorvos
60571	Dieldrin # @
103231	Di(2-ethylhexyl)adipate #
84662	Diethyl phthalate
685916	Diethylacetamide
109897	Diethylamine
110816	Diethyldisulfide
111466	Diethylene glycol
124174	Diethylene glycol butyl ether acetate
108203	Diisopropylether
792745	Dimethyl 4,4'-biphenylcarboxylate
624920	Dimethyl disulfide
80466	Dimethylpropyl phenol
67685	Dimethylsulfoxide
84742	Di- <i>n</i> -butyl phthalate
117840	Di- <i>n</i> -octyl phthalate
84764	Dinonyl phthalate
88857	Dinoseb
101848	Diphenyloxide
2764729	Diquat
115297	Endosulfan
72208	Endrin
64175	Ethanol
637923	Ethyl <i>tert</i> -butyl ether
60297	Ethyl ether
75047	Ethylamine
100414	Ethylbenzene #
107211	Ethylene glycol
103361097	Flumioxazin
206440	Fluoranthene
86737	Fluorene
16984488	Fluoride
59756604	Fluridone
50000	Formaldehyde
64186	Formic acid
25812300	Gemfibrozil
76448	Heptachlor #
1024573	Heptachlor epoxide #
87821	Hexabromobenzene
118741	Hexachlorobenzene # @
87683	Hexachlorobutadiene # @
319846	<i>alpha</i> -Hexachlorocyclohexane # @
319857	<i>beta</i> -Hexachlorocyclohexane # @

319868	<i>delta</i> -Hexachlorocyclohexane @
77474	Hexachlorocyclopentadiene
67721	Hexachloroethane #
2691410	HMX
302012	Hydrazine #
7722841	Hydrogen peroxide
7783064	Hydrogen sulfide
15687271	Ibuprofen
138261413	Imidacloprid
288324	Imidazole
193395	Indeno (1,2,3-cd)pyrene #
64741668	Isopar C
78591	Isophorone #
98828	Isopropylbenzene
99876	<i>p</i> -Isopropyltoluene
7439921	Lead
58899	Lindane # @
7439932	Lithium
7439965	Manganese
950107	Mephosfolan
7439976	Mercury @
79390	Methacrylamide
67561	Methanol
40596698	Methoprene
74931	Methylmercaptan
625456	Methoxyacetic acid
108872	Methylcyclohexane
78933	Methyl ethyl ketone
1634044	Methyl <i>tert</i> butyl ether (MTBE) #
74895	Methylamine
75092	Methylene chloride #
51218452	Metolachlor #
2385855	Mirex # @
7439987	Molybdenum
38836394	N-(1-methylpropylidene)-2-propanamine
140807	N,N-diethyl-1,4-pentanediamine
927628	N,N-dimethyl-1-butamine
127195	N,N-dimethylacetamide
625503	N-ethyl acetamide
103695	N-ethylaniline
100618	N-methylaniline
91203	Naphthalene
22204531	Naproxen
7440020	Nickel
14797558	Nitrate, expressed as nitrogen
98953	Nitrobenzene #
25154523	<i>n</i> -Nonylphenol (mixed isomers)
872504	N-Methyl pyrrolidone
103651	N-Propylbenzene
706785	Octachlorocyclopentene

140669	Octylphenol
144627	Oxalic acid
10028156	Ozone
56382	Parathion
67774327	PBB #
1336363	PCB # @
608935	Pentachlorobenzene # @
87865	Pentachlorophenol #
79210	Peracetic acid
14797730	Perchlorate
375735	Perfluorobutanesulfonate
335671	Perfluorooctanoic acid
1763231	Perfluorooctane sulfonate @
85018	Phenanthrene
108952	Phenol
57410	Phenytoin #
32289580	PHMB
1918021	Picloram
51036	Piperonyl butoxide
7722647	Potassium permanganate
79094	Propionic acid
57556	Propylene glycol
75569	Propylene oxide #
129000	Pyrene
8003347	Pyrethrin
121824	RDX #
7782492	Selenium & inorganic salts
7440224	Silver
93721	Silvex
122349	Simazine
26628228	Sodium azide
140932	Sodium isopropyl xanthate
168316958	Spinosad
7440246	Strontium
100425	Styrene #
14808798	Sulfate
126863	Surfynol 104E
3383968	Temephos
51207319	Tetrachlorodibenzofuran, 2,3,7,8-
127184	Tetrachloroethylene #
109999	Tetrahydrofuran
7440280	Thallium
7440326	Titanium
620235	m-Tolualdehyde
108883	Toluene
8001352	Toxaphene # @
102829	Tributylamine
79016	Trichloroethylene #
3380345	Triclosan
121448	Triethylamine

149735	Trimethylorthoformate
126727	Tris(2,3-dibromopropyl)phosphate #
7440622	Vanadium
75014	Vinyl chloride #
1330207	Xylene
7440666	Zinc

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00000	Formaldehyde	5000 ###	###	NA	NA	NA	180 ###	790 ###	1600 ###
50293	DDT # @	0.002 ###	0.002 ###	1E-005 ###	0.00015 ###	0.0002 ###	0.0032 ###	0.029 ###	0.057 ###
50328	Benzo(a)pyrene #	NLS	NLS	NA	NLS	NLS	2.7 ###	25 ###	49 ###
51036	Piperonyl butoxide	NLS	NLS	NA	NA	NA	0.45 ###	4.0 ###	8.1 ###
51285	2,4-Dinitrophenol	55 ###	2800 ###	NA	NA	NA	19 ###	130 ###	270 ###
53703	Dibenz(a,h)anthracene	NLS	NLS	NA	NLS	NLS	ID	ID	ID
56235	Carbon tetrachloride #	18 ###	140 ###	NA	4.7 ###	38 ###	77 ###	690 ###	1400 ###
56382	Parathion	NLS	NLS	NA	NA	NA	0.013 ###	0.065 ###	0.13 ###
56553	Benzo(a)anthracene	NLS	NLS	NA	NLS	NLS	0.43 ###	3.9 ###	7.8 ###
57125	Cyanide, free	600 ###	48000 ###	NA	NA	NA	5.2 ###	22 ###	44 ###
57410	Phenytain #	790 ###	11000 ###	NA	6.1 ###	89 ###	120 ###	1100 ###	2200 ###
57556	Propylene glycol	###	###	NA	NA	NA	290000 ###	1000000 ###	2100000 ###
57749	Chlordane # @	0.0014 ###	0.0014 ###	NLS	0.00025 ###	0.0003 ###	0.029 ###	0.27 ###	0.53 ###
58899	Lindane # @	0.47 ###	0.5 ###	0.026 ###	0.025 ###	0.027 ###	0.07 ###	0.95 ###	1.9 ###
58902	2,3,4,6-Tetrachloroph	120 ###	150 ###	NA	NA	NA	1.2 ###	11 ###	22 ###
59507	4-Chloro-3-methylphe	6900 ###	39000 ###	NA	NA	NA	7.4 ###	67 ###	130 ###
60297	Ethyl ether	14000 ###	###	NA	NA	NA	830 ###	7500 ###	15000 ###
60571	Dieldrin # @	0.0004 ###	0.0004 ###	7E-005 ###	7E-006 ###	7E-006 ###	0.056 ###	0.24 ###	0.48 ###
62533	Aniline #	190 ###	13000 ###	NA	21 ###	1500 ###	3 ###	10 ###	21 ###
62737	Dichlorvos	NLS	NLS	NA	NA	NA	0.0040 ###	0.038 ###	0.076 ###
63252	Carbaryl	NLS	NLS	NA	NA	NA	0.5200 ###	2.3 ###	4.5 ###
64175	Ethanol	###	###	NA	NA	NA	NLS	NLS	NLS
64186	Formic acid	38000 ###	###	NA	NA	NA	14000 ###	120000 ###	240000 ###
64197	Acetic acid (includes	16000 ###	###	NA	NA	NA	7697.1083627131 ###	69265.345834637 ###	138538.008976086 ###
67561	Methanol	14000 ###	###	NA	NA	NA	590000 ###	1300000 ###	2700000 ###
67630	2-Propanol	28000 ###	###	NA	NA	NA	57000 ###	500000 ###	1000000 ###
67641	Acetone	5600 ###	###	NA	NA	NA	1700 ###	15000 ###	30000 ###
67663	Chloroform #	350 ###	11000 ###	NA	*	*	630 ###	5700 ###	11000 ###
67685	Dimethylsulfoxide	###	###	NA	NA	NA	190000 ###	1700000 ###	3400000 ###
67721	Hexachloroethane #	6 ###	7.6 ###	NA	5.3 ###	6.7 ###	13 ###	110 ###	210 ###
71363	n-Butanol	3500 ###	###	NA	NA	NA	9800 ###	88000 ###	180000 ###
71432	Benzene #	19 ###	510 ###	NA	12 ###	310 ###	200 ###	950 ###	1900 ###
71501	Acetate (includes Acet	16000 ###	###	NA	NA	NA	7697.1083627131 ###	69265.345834637 ###	138538.008976086 ###
71556	1,1,1-Trichloroethane	62000 ###	###	NA	NA	NA	89 ###	800 ###	1600 ###
72208	Endrin	NLS	NLS	NA	NA	NA	0.036 ###	0.086 ###	0.172 ###
74839	Bromomethane	39 ###	2600 ###	NA	NA	NA	4.2 ###	38 ###	75 ###
74873	Chloromethane #	3500 ###	###	NA	110 ###	7300 ###	490 ###	4400 ###	8900 ###
74895	Methylamine	31 ###	2500 ###	NA	NA	NA	41 ###	370 ###	740 ###
74931	Methylmercaptan	140 ###	11000 ###	NA	NA	NA	1200 ###	11000 ###	22000 ###
74975	Bromochloromethane	1000 ###	59000 ###	NA	NA	NA	610 ###	5500 ###	11000 ###
75003	Chloroethane #	###	###	NA	170 ###	9400 ###	1100 ###	10000 ###	20000 ###
75014	Vinyl chloride #	83 ###	4400 ###	NA	0.25 ###	13 ###	930 ###	8400 ###	17000 ###
75047	Ethylamine	740 ###	60000 ###	NA	NA	NA	31 ###	300 ###	610 ###
75058	Acetonitrile	5600 ###	###	NA	NA	NA	13000 ###	120000 ###	230000 ###
75070	Acetaldehyde	1200 ###	93000 ###	NA	NA	NA	130 ###	1200 ###	2400 ###
75092	Methylene chloride #	1600 ###	90000 ###	NA	47 ###	2600 ###	1500 ###	8500 ###	17000 ###
75150	Carbon disulfide	1000 ###	34000 ###	NA	NA	NA	9.0 ###	81 ###	160 ###
75252	Bromoform #	470 ###	8100 ###	NA	52 ###	890 ###	160 ###	1400 ###	2800 ###
75274	Bromodichloromethane	170 ###	4500 ###	NA	6.8 ###	180 ###	180 ###	1700 ###	3300 ###
75343	1,1-Dichloroethane	9800 ###	###	NA	NA	NA	740 ###	6600 ###	13000 ###
75354	1,1-Dichloroethylene	1200 ###	33000 ###	NA	NA	NA	130 ###	1200 ###	2300 ###
75434	Dichlorodifluoromethane	2500 ###	###	NA	NA	NA	150 ###	1300 ###	2600 ###
75569	Propylene oxide #	ID*	ID*	NA	2.3 ###	180 ###	220 ###	2000 ###	4000 ###
75718	Dichlorodifluorometh	2900 ###	90000 ###	NA	NA	NA	170 ###	1500 ###	3000 ###
76131	1,1,2-Trichloro-1,2,2-t	###	###	NA	NA	NA	32 ###	280 ###	570 ###
76448	Heptachlor #	0 ###	0.072 ###	NA	0.0017 ###	0.0018 ###	0.07 ###	0.42 ###	0.85 ###
77474	Hexachlorocyclopenta	140 ###	450 ###	NA	NA	NA	0.07 ###	0.27 ###	0.54 ###
77769	2,2-Dimethoxypropan	ID	ID	NA	NA	NA	6900 ###	62000 ###	120000 ###
78591	Isophorone #	4100 ###	###	NA	310 ###	8200 ###	1300 ###	4600 ###	9200 ###
78875	1,2-Dichloropropane #	12000 ###	###	NA	9.1 ###	290 ###	230 ###	2000 ###	4000 ###
78933	Methyl ethyl ketone	17000 ###	###	NA	NA	NA	2200 ###	20000 ###	40000 ###
79005	1,1,2-Trichloroethane	110 ###	3000 ###	NA	12 ###	330 ###	730 ###	3200 ###	6400 ###
79016	Trichloroethylene #	44 ###	550 ###	NA	29 ###	370 ###	200 ###	1800 ###	3500 ###
79061	Acrylamide #	28 ###	2200 ###	NA	0.12 ###	10 ###	590 ###	5300 ###	11000 ###
79094	Propionic acid	50000 ###	###	NA	NA	NA	3600 ###	33000 ###	65000 ###
79210	Peracetic acid	NLS	NLS	NA	NA	NA	50 ###	110 ###	230 ###
79345	1,1,2,2,2-Tetrachloroeth	180 ###	4300 ###	NA	3.2 ###	78 ###	200 ###	910 ###	1800 ###
79390	Methacrylamide	NLS	NLS	NLS	NLS	NLS	4300 ###	38000 ###	77000 ###
80057	Bisphenol A	NLS	NLS	NA	NA	NA	20 ###	180 ###	350 ###

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Rule 57 Water Quality Values
Surface Water Assessment Section
EGL
July 22, 2022
DRAFT

* See criteria files		HNV		HNV		WV		HCV		HCV		FCV		AMV		FAV	
CAS #	PARAMETER NAME	Drink	Non-drink	Drink	Non-drink	Drink	Non-drink	Drink	Non-drink	Drink	Non-drink	Value	Verif	Value	Verif	Value	Verif
		Value	date	Value	date	Value	date	Value	date	Value	date	Value	date	Value	date	Value	date
80466	Dimethylpropyl pheno	26	###	38	###	NA		NA		NA		5.8	###	52	###	100	###
83329	Acenaphthene	580	###	890	###	NA		NA		NA		38	###	100	###	200	###
84662	Diethyl phthalate	14000	###	40000	###	NA		NA		NA		110	###	980	###	2000	###

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Rule 57 Water Quality Values
Surface Water Assessment Section
EGLE
July 22, 2022
DRAFT

CAS #	PARAMETER NAME	HNV Drink Value verif date	HNV Non-drink Value verif date	WV Value verif date	HCV Drink Value verif date	HCV Non-drink Value verif date	FCV Value verif date	AMV Value verif date	FAV Value verif date
84742	Di-n-butyl phthalate	640 ###	690 ###	NA	NA	NA	9.7 ###	38 ###	75 ###
84764	Dinonyl phthalate	ID* ###	ID* ###	NA	NA	NA	140 ###	1200 ###	2500 ###
85018	Phenanthrene	ID* ###	ID* ###	NA	NA	NA	1.7 ###	5.7 ###	11 ###
85687	Butyl benzyl phthalate	6.9 ###	160 ###	NA	NA	NA	67 ###	310 ###	630 ###
86737	Fluorene	140 ###	160 ###	NA	NA	NA	12 ###	110 ###	220 ###
86748	Carbazole #	ID* ###	ID* ###	NA	19 ###	41 ###	4 ###	36 ###	72 ###
87616	1,2,3-Trichlorobenzene	55 ###	73 ###	NA	NA	NA	4.5 ###	40 ###	81 ###
87683	Hexachlorobutadiene	0.093 ###	0.098 ###	0.053 ###	0.33 ###	0.35 ###	1 ###	7 ###	15 ###
87821	Hexabromobenzene	81 ###	6500 ###	NA	NA	NA	ID ###	ID ###	ID ###
87865	Pentachlorophenol #	300 ###	450 ###	NA	1.8 ###	2.8 ###	6.6925836808 ###	8.7233208775 ###	17.446641755 ###
88062	2,4,6-Trichlorophenol	1900 ###	14000 ###	NA	41 ###	290 ###	5 ###	39 ###	79 ###
88697	2-Isopropylphenol	23 ###	380 ###	NA	NA	NA	36 ###	320 ###	650 ###
88755	2-Nitrophenol	78 ###	3100 ###	NA	NA	NA	56 ###	510 ###	1000 ###
88857	Dinoseb	28 ###	1900 ###	NA	NA	NA	0.48 ###	4.8 ###	9.5 ###
90120	1-Methylnaphthalene	740 ###	1200 ###	NA	34 ###	54 ###	9.5 ###	86 ###	170 ###
91203	Naphthalene	1100 ###	2300 ###	NA	NA	NA	11 ###	100 ###	200 ###
91576	2-Methylnaphthalene	600 ###	1000 ###	NA	NA	NA	19 ###	170 ###	340 ###
91941	3,3'-Dichlorobenzidine	650 ###	950 ###	NA	0.14 ###	0.2 ###	4.5 ###	41 ###	81 ###
92524	Biphenyl	460 ###	690 ###	NA	NA	NA	13 ###	54 ###	110 ###
92875	Benzidine #	74 ###	3700 ###	NA	0.0015 ###	0.073 ###	2.7 ###	25 ###	49 ###
93721	Silvex	83 ###	140 ###	NA	NA	NA	30 ###	270 ###	540 ###
93765	2,4,5-T	490 ###	1200 ###	NA	NA	NA	14 ###	130 ###	260 ###
94757	2,4-D	240 ###	1900 ###	NA	NA	NA	220 ###	1400 ###	2900 ###
95487	2-Methylphenol	1400 ###	44000 ###	NA	NA	NA	76 ###	690 ###	1400 ###
95498	o-Chlorotoluene	360 ###	970 ###	NA	NA	NA	9.8 ###	88 ###	180 ###
95501	1,2-Dichlorobenzene	2000 ###	11000 ###	NA	NA	NA	13 ###	120 ###	240 ###
95512	2-Chloroaniline	91 ###	3100 ###	NA	NA	NA	24 ###	210 ###	430 ###
95578	2-Chlorophenol	190 ###	400 ###	NA	NA	NA	18 ###	160 ###	320 ###
95636	1,2,4-Trimethylbenzene	190 ###	330 ###	NA	NA	NA	17 ###	150 ###	310 ###
95658	3,4-Dimethylphenol	37 ###	580 ###	NA	NA	NA	25 ###	230 ###	460 ###
95943	1,2,4,5 - Tetrachloro	2.8 ###	2.9 ###	NA	NA	NA	3 ###	23 ###	46 ###
96128	1,2-Dibromo-3-chloro	ID* ###	ID* ###	NA	0.24 ###	4.9 ###	75 ###	670 ###	1300 ###
98066	tert-Butylbenzene	82 ###	110 ###	NA	NA	NA	ID ###	ID ###	ID ###
98668	4-Chlorobenzene sulfide	28000 ###	###	NA	NA	NA	1300 ###	150000 ###	290000 ###
98828	Isopropylbenzene	1700 ###	3800 ###	NA	NA	NA	28 ###	250 ###	500 ###
98862	Acetophenone	5700 ###	###	NA	NA	NA	450 ###	4100 ###	8200 ###
98953	Nitrobenzene #	26 ###	990 ###	NA	4.7 ###	180 ###	230 ###	1000 ###	2100 ###
99876	p-Isopropyltoluene	33 ###	45 ###	NA	NA	NA	3.2 ###	29 ###	58 ###
99898	4-Isopropylphenol	21 ###	130 ###	NA	NA	NA	20 ###	180 ###	360 ###
###	4-Nitrophenol	680 ###	18000 ###	NA	NA	NA	200 ###	940 ###	1900 ###
###	Ethylbenzene #	1600 ###	6900 ###	NA	25 ###	110 ###	21 ###	190 ###	370 ###
###	Styrene #	4200 ###	18000 ###	NA	20 ###	80 ###	160 ###	1400 ###	2900 ###
###	N-Methylaniline	ID ###	ID ###	NA	NA	NA	360 ###	3200 ###	6,500 ###
###	Diphenyloxide	44 ###	78 ###	NA	NA	NA	2.9 ###	26 ###	52 ###
###	Tributylamine	ID ###	ID ###	NA	NA	NA	2.8 ###	25 ###	50 ###
###	Di(2-ethylhexyl)adipate	3500 ###	6500 ###	NA	150 ###	280 ###	4.6 ###	41 ###	83 ###
###	Azobenzene #	ID* ###	ID* ###	NA	3.7 ###	6.0 ###	4.1 ###	37 ###	74 ###
###	N-Propylbenzene	150 ###	290 ###	NA	NA	NA	4.9 ###	44 ###	89 ###
###	N-Ethylaniline	7.8 ###	190 ###	NA	NA	NA	1.8 ###	16 ###	32 ###
###	n-Butylbenzene	48 ###	57 ###	NA	NA	NA	2.7 ###	24 ###	48 ###
###	2-Ethylhexanol #	8700 ###	68000 ###	NA	86 ###	660 ###	130 ###	1100 ###	2300 ###
###	2,4-Dimethylphenol	450 ###	8700 ###	NA	NA	NA	380 ###	1300 ###	2700 ###
###	4-Methylphenol	1400 ###	45000 ###	NA	NA	NA	25 ###	230 ###	450 ###
###	1,4-Dichlorobenzene #	1100 ###	11000 ###	NA	24 ###	240 ###	17 ###	100 ###	210 ###
###	4-Chloroaniline #	39 ###	1400 ###	NA	2 ###	72 ###	19 ###	170 ###	340 ###
###	4-Chlorophenol	880 ###	14000 ###	NA	NA	NA	30 ###	270 ###	530 ###
###	1,2-Dibromoethane #	250 ###	8200 ###	NA	0.17 ###	5.7 ###	15 ###	140 ###	280 ###
###	1,2-Dichloroethane #	6900 ###	###	NA	6.0 ###	360 ###	2000 ###	8200 ###	16000 ###
###	Acrylonitrile #	58 ###	320 ###	NA	0.21 ###	1.2 ###	66 ###	590 ###	1200 ###
###	Ethylene glycol	56000 ###	###	NA	NA	NA	190000 ###	1700000 ###	3400000 ###
###	4-Methyl-2-pentanone	1400 ###	85000 ###	NA	NA	NA	1000 ###	6700 ###	13000 ###
###	4-Methyl-2-pentanol	1400 ###	85000 ###	NA	NA	NA	320 ###	2800 ###	5900 ###
###	Diisopropylether	7300 ###	###	NA	NA	NA	470 ###	4200 ###	8400 ###
###	3-Methylphenol	2700 ###	89000 ###	NA	NA	NA	71 ###	636 ###	1271 ###
###	Bis(2-chloroisopropyl)	990 ###	47000 ###	NA	6 ###	290 ###	540 ###	4900 ###	9700 ###
###	1,3,5-Trimethylbenzene	2000 ###	4200 ###	NA	NA	NA	45 ###	410 ###	810 ###
###	Methylcyclohexane	34 ###	54 ###	NA	NA	NA	3.1 ###	28 ###	55 ###
###	Toluene	5600 ###	51000 ###	NA	NA	NA	260 ###	1300 ###	2600 ###
###	Chlorobenzene	470 ###	3200 ###	NA	NA	NA	25 ###	220 ###	450 ###

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Rule 57 Water Quality Values

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EGLE

July 22, 2022

DRAFT

* See criteria files

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###	Phenol	1100 ###	1200 ###	NA	NA	NA	450 ###	3,400 ###	6,800 ###
###	Butylamine	45 ###	3100 ###	NA	NA	NA	57 ###	510 ###	1000 ###

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EGLE
July 22, 2022
DRAFT

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CAS #	PARAMETER NAME	Value	Value	Value	Value	Value	Value	Value	Value
###	Diethylamine	310 ###	24000 ###	NA	NA	NA	20 ###	180 ###	350 ###
###	Tetrahydrofuran	350 ###	26000 ###	NA	NA	NA	11000 ###	74000 ###	150000 ###
###	Diethyldisulfide	50 ###	380 ###	NA	NA	NA	62 ###	560 ###	1100 ###
###	Cyclohexane	84000 ###	###	NA	NA	NA	15 ###	140 ###	280 ###
###	Bis(2-chloroethyl)ether	ID* ###	ID* ###	NA	0.79 ###	15 ###	ID*	9200 ###	18000 ###
###	Diethylene glycol	###	###	NA	NA	NA	43000 ###	380000 ###	770000 ###
###	Dibutylamine	6.7 ###	69 ###	NA	NA	NA	36 ###	320 ###	650 ###
###	1,2-Bis(2-chloroethoxy)ethane	6.9 ###	460 ###	NA	NA	NA	1500 ###	13000 ###	26000 ###
###	Endosulfan	85 ###	170 ###	NA	NA	NA	0.029 ###	0.13 ###	0.27 ###
###	Bis(2-ethylhexyl)phthalate	120 ###	160 ###	NA	14 ###	18 ###	ID*	285 ###	285 ###
###	Di-n-octyl phthalate	240 ###	300 ###	NA	NA	NA	ID	ID	ID
###	Hexachlorobenzene #	0.046 ###	0.046 ###	0.0003 ###	0.00045 ###	0.0005 ###	ID*	610 ###	1200 ###
###	Anthracene	1900 ###	2400 ###	NA	NA	NA	3.2 ###	29 ###	58 ###
###	1,2,4-Trichlorobenzene	80 ###	99 ###	NA	NA	NA	130 ###	420 ###	850 ###
###	2,4-Dichlorophenol	220 ###	1100 ###	NA	NA	NA	11 ###	92 ###	180 ###
###	2,4-Di-tert-pentylphenol	0.58 ###	0.61 ###	NA	NA	NA	NLS	NLS	NLS
###	Triethylamine	4000 ###	###	NA	NA	NA	260 ###	1100 ###	2100 ###
###	RDX #	83 ###	6100 ###	NA	5.8 ###	420 ###	85 ###	400 ###	790 ###
###	Simazine	140 ###	4000 ###	NA	NA	NA	17 ###	160 ###	310 ###
###	1,4-Dioxane #	890 ###	72000 ###	NA	3.5 ###	280 ###	22000 ###	200000 ###	390000 ###
###	Diethylene glycol butyl ether	330 ###	13000 ###	NA	NA	NA	260 ###	2300 ###	4600 ###
###	Dibromochloromethane	570 ###	12000 ###	NA	6.8 ###	150 ###	170 ###	1600 ###	3100 ###
###	Tris(2,3-dibromopropyl)phosphate	ID* ###	ID* ###	NA	0.31 ###	13 ###	1.4 ###	12 ###	25 ###
###	Tetramethyl decynedide	ID* ###	ID* ###	NA	NA	NA	350 ###	3,100 ###	6200 ###
###	Tetrachloroethylene #	60 ###	330 ###	NA	11 ###	60 ###	190 ###	1400 ###	2900 ###
###	N,N-Dimethylacetamide	700 ###	57000 ###	NA	NA	NA	4100 ###	37000 ###	74000 ###
###	4-Methyl-2,6-di-T-butylphenol	3 ###	3 ###	NA	0.25 ###	0.25 ###	0.37 ###	3.3 ###	6.6 ###
###	Pyrene	15 ###	15 ###	NA	NA	NA	0.58 ###	5.2 ###	10 ###
###	Dibenzofuran	ID* ###	ID* ###	NA	###	###	4 ###	36 ###	72 ###
###	sec-Butylbenzene	77 ###	100 ###	NA	NA	NA	3.1 ###	28 ###	56 ###
###	Octylphenol	NLS	NLS	NA	NA	NA	2 ###	13 ###	26 ###
###	N,N-diethyl-1,4-pentadiene	5 ###	360 ###	NA	NA	NA	2600 ###	23000 ###	46000 ###
###	Sodium isopropyl xanthate	22 ###	1800 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	Oxalic acid	4500 ###	###	NA	NA	NA	250 ###	2300 ###	4600 ###
###	Trimethylorthoformate	87 ###	6700 ###	NA	NA	NA	1600 ###	15000 ###	29000 ###
###	cis-1,2-Dichloroethylene	880 ###	36000 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	trans-1,2-Dichloroethylene	470 ###	19000 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	Benzo(g,h,i)perylene	17 ###	19 ###	NA	NA	NA	3 ###	27 ###	54 ###
###	Indeno (1,2,3-cd)pyrene	NLS	NLS	NA	NLS	NLS	ID	ID	ID
###	Benzo(b)fluoranthene	NLS	NLS	NA	NLS	NLS	ID	ID	ID
###	Fluoranthene	18 ###	18 ###	NA	NA	NA	1.6 ###	14 ###	28 ###
###	Acenaphthylene	260 ###	530 ###	NA	NA	NA	6.6 ###	60 ###	120 ###
###	Chrysene #	ID* ###	ID* ###	NA	1.5 ###	1.5 ###	0.43 ###	3.9 ###	7.8 ###
###	Imidazole	27 ###	2200 ###	NA	NA	NA	1500 ###	13000 ###	26000 ###
###	Hydrazine #	42 ###	3400 ###	NA	0.094 ###	7.6 ###	1.8 ###	16 ###	32 ###
###	Aldrin #	0.0001 ###	0.0001 ###	NA	9E-006 ###	9E-006 ###	0.017 ###	0.15 ###	0.3 ###
###	alpha-Hexachlorocyclopentadiene	19 ###	21 ###	ID*	0.016 ###	0.017 ###	ID	ID	ID
###	beta-Hexachlorocyclopentadiene	0.091 ###	0.098 ###	ID*	0.024 ###	0.026 ###	ID	ID	ID
###	delta-Hexachlorocyclopentadiene	28 ###	30 ###	ID	NA	NA	ID	ID	ID
###	Diazinon	12 ###	37 ###	NA	NA	NA	0.004 ###	0.064 ###	0.13 ###
###	Perfluorooctanoic acid	0.066 ###	0.17 ###	NA	NA	NA	880 ###	7700 ###	15000 ###
###	Perfluorobutanesulfonic acid	8.3 ###	670 ###	NA	NA	NA	24000 ###	120000 ###	240000 ###
###	Azoxbenzene	17 ###	1400 ###	NA	NA	NA	ID	ID	ID
###	1,2,3-Trimethylbenzene	290 ###	650 ###	NA	NA	NA	6.5 ###	59 ###	120 ###
###	2,3-Dimethylphenol	16 ###	270 ###	NA	NA	NA	120 ###	1100 ###	2200 ###
###	1,2-Dichloroethylene	470 ###	19000 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	1,3-Dichlorobenzene	37 ###	65 ###	NA	NA	NA	28 ###	100 ###	200 ###
###	1,3-Dichloropropylene	930 ###	39000 ###	NA	3.3 ###	140 ###	9 ###	81 ###	160 ###
###	1,7-Dimethylnaphthalene	ID	ID	NA	NA	NA	2.8 ###	26 ###	51 ###
###	3-tert-Butylphenol	110 ###	360 ###	NA	NA	NA	29 ###	260 ###	530 ###
###	2-Hexanone	9700 ###	###	NA	NA	NA	570 ###	5200 ###	10000 ###
###	2,2-Dichloropropane	ID	ID	NA	NA	NA	80 ###	720 ###	1400 ###
###	Pentachlorobenzene #	0.38 ###	0.38 ###	0.019 ###	NA	NA	1.2 ###	11 ###	21 ###
###	2-Ethyltoluene	71 ###	150 ###	NA	NA	NA	ID	ID	ID
###	3-Isopropylphenol	39 ###	250 ###	NA	NA	NA	26 ###	240 ###	470 ###
###	3-Ethyltoluene	67 ###	130 ###	NA	NA	NA	ID	ID	ID
###	m-Tolualdehyde	ID*	ID*	NA	NA	NA	34 ###	310 ###	620 ###
###	Dimethyl disulfide	480 ###	20000 ###	NA	NA	NA	ID	ID	ID
###	Methoxyacetic acid	28 ###	2200 ###	NA	NA	NA	270 ###	2400 ###	4800 ###

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EGLE

July 22, 2022

DRAFT

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###	N-Ethyl acetamide	87 ###	7000 ###	NA	NA	NA	6900 ###	62000 ###	120000 ###
###	1,1,1,2-Tetrachloroeth	680 ###	3500 ###	NA	19 ###	100 ###	85 ###	770 ###	1500 ###
###	1,1,3,3-Tetramethylur	39 ###	3100 ###	NA	NA	NA	7700 ###	69000 ###	140000 ###
###	1,2,3,4-Tetrachlorobe	3.9 ###	4.1 ###	ID ###	NA	NA	3.4 ###	18 ###	35 ###
###	Ethyl <i>tert</i> -butyl ether	2500 ###	### ###	NA	NA	NA	350 ###	3100 ###	6300 ###
###	Diethylacetamide	44 ###	3400 ###	NA	NA	NA	6400 ###	58000 ###	120000 ###
###	Octachlorocyclopente	0.014 ###	0.014 ###	NA	NA	NA	0.08 ###	0.75 ###	1.5 ###
###	Dimethyl 4,4'-bipheny	ID ###	ID ###	NA	NA	NA	ID ###	ID ###	ID ###
###	N-Methyl pyrrolidone	25000 ###	### ###	NA	NA	NA	6600 ###	59000 ###	120000 ###

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Rule 57 Water Quality Values
Surface Water Assessment Section
EGLE
July 22, 2022
DRAFT

* See criteria files

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###	N,N-dimethyl-1-butan	ID ###	ID ###	NA	NA	NA	510 ###	4600 ###	9200 ###
###	Mephosfolan	2.5 ###	170 ###	NA	NA	NA	0.37 ###	3.3 ###	6.6 ###
###	Heptachlor epoxide #	0.0075 ###	0.0076 ###	NA	0.0021 ###	0.0021 ###	0.037 ###	0.33 ###	0.66 ###
###	Xylene	3800 ###	16000 ###	NA	NA	NA	49 ###	440 ###	890 ###
###	PCB # @	NLS	NLS	0.0001 ###	3E-005 ###	3E-005 ###	ID* ###	ID ###	ID ###
###	Methyl tert butyl ethe	920 ###	63000 ###	NA	100 ###	7100 ###	32000 ###	210000 ###	420000 ###
###	3,6-Dichloropicolinic	4100 ###	###	NA	NA	NA	660 ###	5900 ###	12000 ###
###	2,3,7,8-TCDD # @	###	###	3E-009 ###	8.6E-009 ###	9E-009 ###	ID* ###	ID* ###	ID* ###
###	Perfluorooctane sulfo	0.011 ###	0.012 ###	NA	NA	NA	140 ###	780 ###	1600 ###
###	Atrazine	880 ###	8600 ###	NA	NA	NA	7.3 ###	50 ###	100 ###
###	Picloram	5500 ###	###	NA	NA	NA	46 ###	290 ###	570 ###
###	Mirex # @	*	*	2E-005 ###	4E-005 ###	4E-005 ###	*	ID ###	ID ###
###	HMX	1400 ###	###	NA	NA	NA	250 ###	2300 ###	4600 ###
###	Diquat	61 ###	4900 ###	NA	NA	NA	6 ###	54 ###	110 ###
###	Chlorpyrifos	*	*	NA	NA	NA	0.0020 ###	0.027 ###	0.053 ###
###	Triclosan	NLS	NLS	NA	NA	NA	4.4000 ###	25 ###	50 ###
###	Temephos	NLS	NLS	NA	NA	NA	0.0001 ###	0.00067 ###	0.0013 ###
###	6-Chloropicolinic acid	410 ###	22000 ###	NA	NA	NA	26 ###	230 ###	460 ###
###	1-Chlorohexadecane	ID ###	ID ###	NA	NA	NA	ID ###	ID ###	ID ###
###	Lead	14 ###	190 ###	NA	NA	NA	18.5613506884 ###	90.6477846926 ###	181.2870159605 ###
###	Lithium	720 ###	58000 ###	NA	NA	NA	440 ###	910 ###	1800 ###
###	Manganese	1300 ###	59000 ###	NA	NA	NA	1437.0938887969 ###	3100.6846490359 ###	6201.0767208978 ###
###	Mercury @	0.0018 ###	0.0018 ###	0.0013 ###	NA	NA	0.77 ###	1.4 ###	2.8 ###
###	Molybdenum	120 ###	10000 ###	NA	NA	NA	3200 ###	29000 ###	58000 ###
###	Nickel	2600 ###	###	NA	NA	NA	38.4601432866 ###	346.272131138 ###	692.5442622761 ###
###	Silver	130 ###	11000 ###	NA	NA	NA	0.06 ###	0.54 ###	1.1 ###
###	Strontium	ID* ###	ID* ###	NA	NA	NA	36000 ###	40000 ###	81000 ###
###	Thallium	1.2 ###	3.7 ###	NA	NA	NA	7.2 ###	47 ###	94 ###
###	Titanium	NLS	NLS	NA	NA	NA	ID ###	ID ###	ID ###
###	Antimony	1.7 ###	130 ###	NA	NA	NA	240 ###	1100 ###	2300 ###
###	Arsenic #	10 ###	280 ###	NA	10 ###	10 ###	150 ###	340 ###	680 ###
###	Barium	5800 ###	###	NA	NA	NA	301.4263153525 ###	854.5932226766 ###	1709.2767260084 ###
###	Beryllium	160 ###	1200 ###	NA	NA	NA	3.6566270011 ###	32.9088341882 ###	65.8145631353 ###
###	Boron	4000 ###	###	NA	NA	NA	7200 ###	34000 ###	69000 ###
###	Cadmium	2.5 ###	130 ###	NA	NA	NA	1.7190854143 ###	2.8969504343 ###	5.7939008687 ###
###	Chromium	120 ###	9400 ###	NA	NA	NA	55.3399302243 ###	425.4317509466 ###	850.8635018933 ###
###	Cobalt	ID* ###	ID* ###	NA	NA	NA	100 ###	370 ###	740 ###
###	Copper	470 ###	38000 ###	NA	NA	NA	6.6029544302 ###	9.6033329695 ###	19.2066659389 ###
###	Vanadium	53 ###	540 ###	NA	NA	NA	27 ###	79 ###	160 ###
###	Zinc	3300 ###	16000 ###	NA	NA	NA	87.3262490635 ###	86.6177196593 ###	173.2354393186 ###
###	Ammonia, total ammo	ID* ###	ID* ###	NA	NA	NA	'see equation in foo ###	'see equation in foo ###	'see equation in footn ###
###	Potassium permangan	NLS	NLS	NA	NA	NA	* ###	29 ###	58 ###
###	Hydrogen peroxide	ID* ###	ID* ###	NA	NA	NA	10 ###	92 ###	180 ###
###	Bromine	ID* ###	ID* ###	NA	NA	NA	0.26 ###	2.4 ###	4.8 ###
###	Selenium & inorganic	120 ###	2700 ###	NA	NA	NA	5 ###	62 ###	120 ###
###	Chlorine	NLS	NLS	NA	NA	NA	* ###	19 ###	38 ###
###	Hydrogen sulfide	160 ###	13000 ###	NA	NA	NA	0.36 ###	3.2 ###	6.4 ###
###	Toxaphene # @	0.021 ###	0.021 ###	0.0001 ###	7E-005 ###	7E-005 ###	0.005 ###	0.15 ###	0.3 ###
###	Pyrethrin	NLS	NLS	NA	NA	NA	0.022 ###	0.2 ###	0.39 ###
###	Ozone	NLS	NLS	NLS	NLS	NLS	0.065 ###	0.58 ###	1.2 ###
###	trans-1,3-Dichloropro	930 ###	39000 ###	NA	3.3 ###	140 ###	9.0 ###	81 ###	160 ###
###	cis-1,3-Dichloropropy	930 ###	39000 ###	NA	3.3 ###	140 ###	9.0 ###	81 ###	160 ###
###	DBNPA	ID* ###	ID* ###	NA	NA	NA	7.8 ###	71 ###	140 ###
###	Nitrate, expressed as n	10000 ###	NLS	NA	NA	NA	NLS	NLS	NLS
###	Perchlorate	19 ###	1600 ###	NA	NA	NA	400 ###	6300 ###	13000 ###
###	Sulfate	ID* ###	ID* ###	NA	NA	NA	370000 ###	600000 ###	1200000 ###
###	Chlorite	830 ###	67000 ###	NA	NA	NA	0.72 ###	6.5 ###	13 ###
###	Bromate #	100 ###	8200 ###	NA	0.5 ###	40 ###	760 ###	6900 ###	14000 ###
###	Ibuprofen	NLS	NLS	NA	NA	NA	96 ###	860 ###	1700 ###
###	Alachlor #	270 ###	6900 ###	NA	3.5 ###	91 ###	11 ###	150 ###	300 ###
###	Chloride	ID* ###	ID* ###	NA	NA	NA	150000 ###	320000 ###	640000 ###
###	Fluoride	NLS	NLS	NA	NA	NA	2113.7792857339 ###	7739.0386619544 ###	15478.8948888776 ###
###	Chromium, hexavalen	120 ###	9400 ###	NA	NA	NA	11 ###	16 ###	32 ###
###	Cyanazine #	190 ###	12000 ###	NA	0.93 ###	56 ###	110 ###	1000 ###	2000 ###
###	Naproxen	NLS	NLS	NA	NLS	NLS	96 ###	860 ###	1700 ###
###	n-Nonylphenol (mixe	NLS	NLS	NA	NA	NA	2 ###	7 ###	14 ###
###	Gemfibrozil	NLS	NLS	NA	NA	NA	44 ###	400 ###	800 ###
###	Sodium azide	330 ###	27000 ###	NA	NA	NA	7.3 ###	65 ###	130 ###

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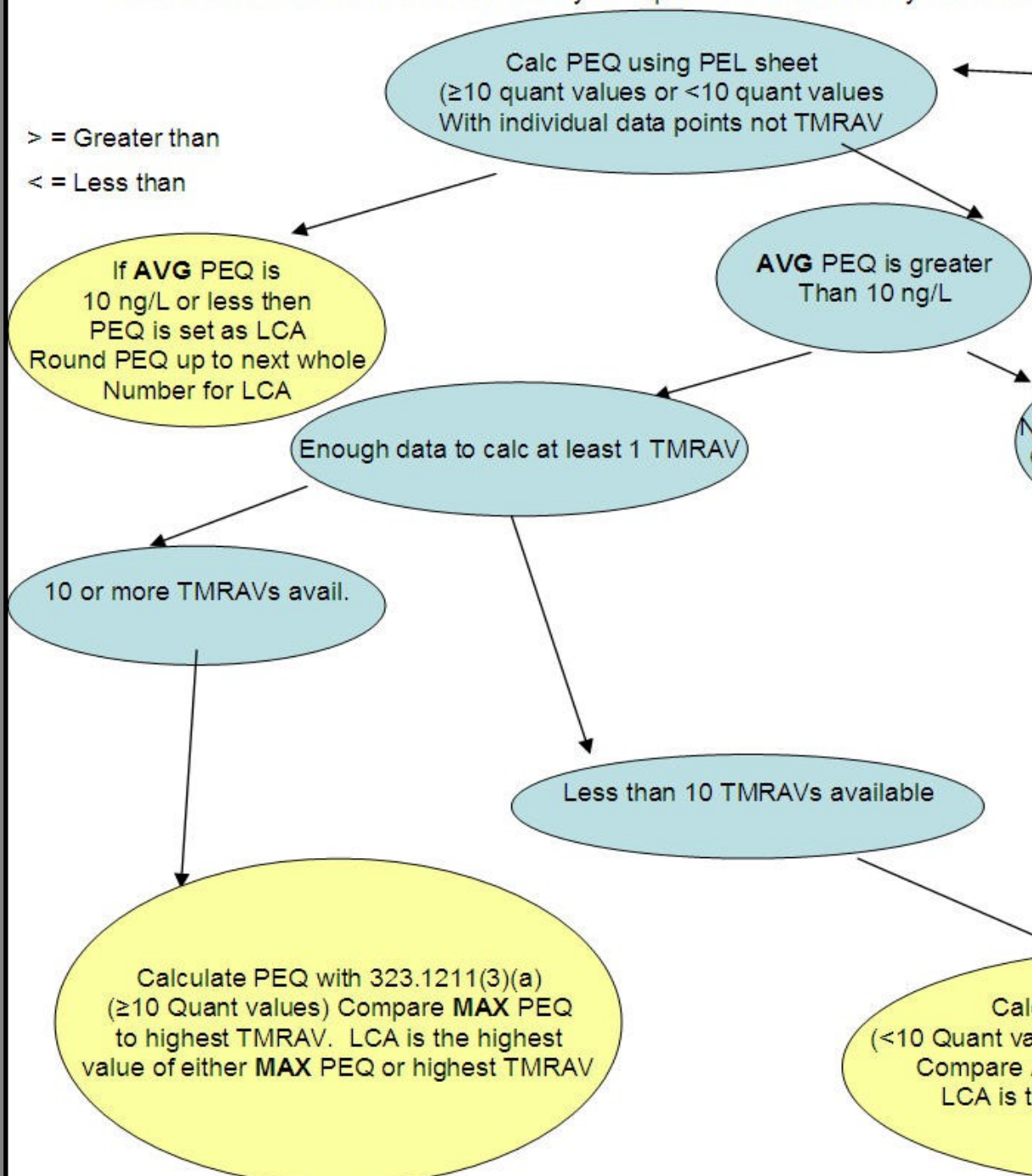
Rule 57 Water Quality Values
Surface Water Assessment Section
EGLE
July 22, 2022
DRAFT

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CAS #	PARAMETER NAME								
###	PHMB	10000 ###	###	NA	51 ###	4100 ###	0.2 ###	1.8 ###	3.6 ###
###	N-(1-methylpropylide	ID ###	ID ###	NA	NA	NA	260 ###	2400 ###	4700 ###
###	3,5,6-Trichloropicolin	ID	ID	NA	NA	NA	ID	ID	ID
###	Methoprene	21 ###	21 ###	NA	NA	NA	2.9 ###	26 ###	51 ###
###	Tetrachlorodibenzofur	NLS	NLS	NA	NA	NA	ID	ID	ID
###	Metolachlor #	3300 ###	14000 ###	NA	78 ###	340 ###	15 ###	110 ###	210 ###
###	Fluridone	2200 ###	80000 ###	NA	NA	NA	1 ###	3 ###	250 ###
###	Isopar C	4500 ###	5600 ###	NA	NA	NA	1.9 ###	17 ###	35 ###
###	PBB #	0.0003 ###	0.0003 ###	NA	0.00013 ###	0.0001 ###	NLS	NLS	NLS
###	4-Nonylphenol	130 ###	200 ###	NA	NA	NA	ID	ID	ID
###	Flumioxazin	NLS	NLS	NA	NA	NA	16 ###	140 ###	290 ###
###	Imidacloprid	NLS	NLS	NA	NA	NA	0.029 ###	0.36 ###	0.72 ###
###	Spinosad	620 ###	7500 ###	NA	NA	NA	30 ###	250 ###	500 ###
###	Cyprosulfamide	11000 ###	###	NA	NA	NA	680 ###	6300 ###	13000 ###
1 Ammonia Aquatic Life Equations (mg/L total ammonia nitrogen; TAN) T = Temperature (in °C) MAX = Maximum value in following parentheses $4 / (1 + 10^{(7.204 - \text{pH})}) + 1.6181 / (1 + 10^{(\text{pH} - 7.204)}) * (14.70 * 10^{(0.036 * (20 - \text{MAX}(T, 13)))})$ $4 / (1 + 10^{(7.204 - \text{pH})}) + 1.6181 / (1 + 10^{(\text{pH} - 7.204)}) * (14.70 * 10^{(0.036 * (20 - \text{MAX}(T, 13)))})$ $(1 + 10^{(7.688 - \text{pH})}) + 1.1994 / (1 + 10^{(\text{pH} - 7.688)}) * (2.276 * 10^{(0.028 * (20 - \text{MAX}(T, 7)))})$									

Flow chart for the calculation of Mercury site specific Levels currently achievable

> = Greater than

< = Less than



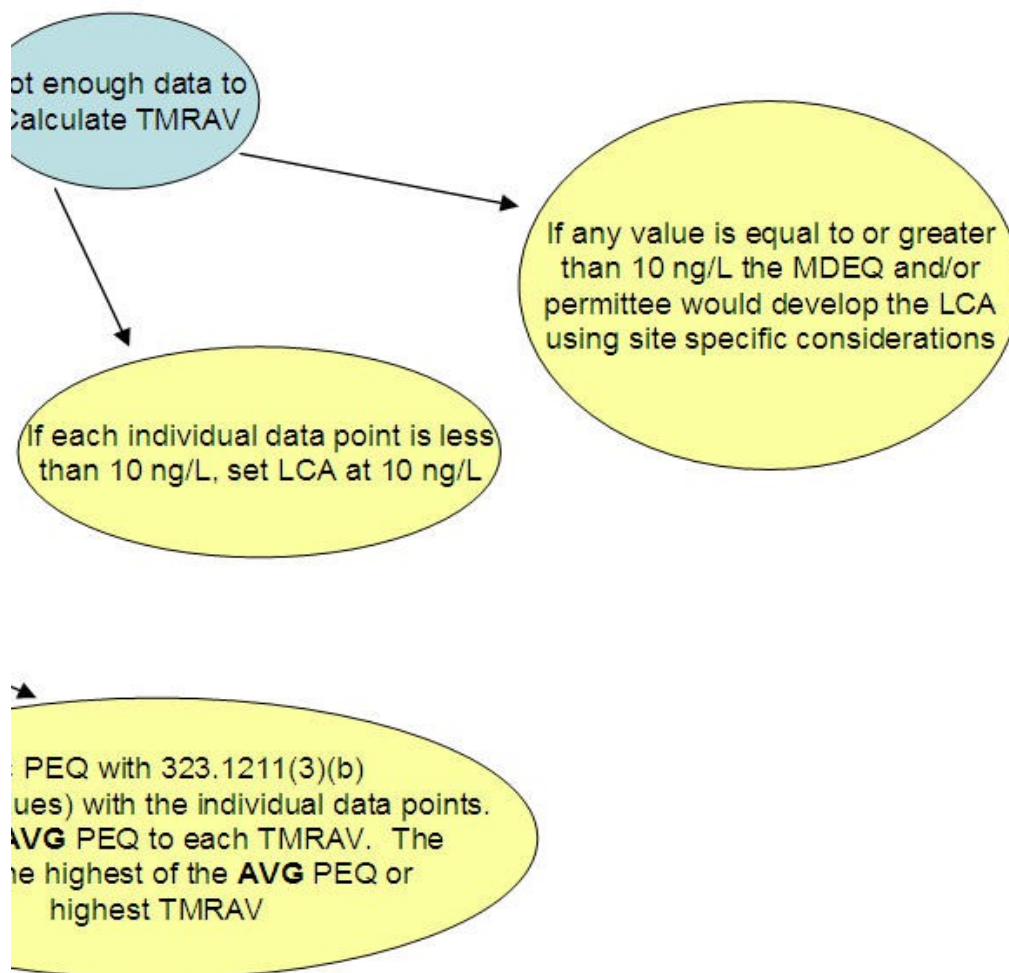
[Click here for full text of Water Bureau](#)

3 (LCA's)

TMRAV=12 month rolling avg.

Start Here

The yellow bubbles are endpoints for the
Calculation of site specific LCAs



[au LCA procedure](#)

REASONABLE POTENTIAL ANALYSIS: WET

Facility: Cadillac WWTP
Outfall: 001

Qe	Qr	RWC	PEL		PEQ*	
			TU _c	TU _A	TU _c	TU _A
4.9535603715	0.65	88.40	1.13121875	1	0.00	0.00

* PEQ (enter data for most sensitive species only):

Chronic

Total n = Quant n = CV =
TU_c effluent = MF = PEQ TU_c = 0.00

Acute

Total n = Quant n = CV =
TU_A effluent = MF = PEQ TU_A = 0.00

Decision (WET limits/conditions):

WQBEL		Species	TI/RE	Monitoring	
Monthly Avg	Daily Max			Short term	Long term

Basis of decision:

WET Data Summary

Modified 10/4/00

EGLE
MI0020257 v7.0
Approved
Issued On:11/29/2023
Expires On:10/01/2027

Species 1: _____

Lab	Test Date	TU		Reference
		Acute	Chronic	

Species 2: _____

Lab	Test Date	TU		Reference
		Acute	Chronic	

Reasonable Potential Multiplying Factors: 95% Confidence Level and 95% Probability Basis

n	Coefficient of Variation																			
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	-	-	-	-	-	6.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	3.8	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	2.6	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	-	-	-	-	-	2.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	-	-	-	-	-	2.1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	-	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
8	-	-	-	-	-	1.9	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	-	-	-	-	-	1.8	-	-	-	-	-	-	-	-	-	-	-	-	-	-
10	1.1	1.2	1.3	1.5	1.6	1.7	1.9	2	2.2	2.3	2.4	2.6	2.7	2.8	3	3.1	3.2	3.3	3.4	3.6
11	1.1	1.2	1.3	1.4	1.6	1.7	1.8	1.9	2.1	2.2	2.3	2.4	2.5	2.7	2.8	2.9	3	3.1	3.2	3.3
12	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.9	2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3	3
13	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2	2.1	2.2	2.3	2.4	2.5	2.5	2.6	2.7	2.8	2.9
14	1.1	1.2	1.3	1.4	1.4	1.5	1.6	1.7	1.8	1.9	2	2.1	2.2	2.3	2.3	2.4	2.5	2.6	2.6	2.7
15	1.1	1.2	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.8	1.9	2	2.1	2.2	2.2	2.3	2.4	2.4	2.5	2.5
16	1.1	1.1	1.2	1.3	1.4	1.5	1.6	1.6	1.7	1.8	1.9	1.9	2	2.1	2.1	2.2	2.3	2.3	2.4	2.4
17	1.1	1.1	1.2	1.3	1.4	1.4	1.5	1.6	1.7	1.7	1.8	1.9	1.9	2	2	2.1	2.2	2.2	2.3	2.3
18	1.1	1.1	1.2	1.3	1.3	1.4	1.5	1.6	1.6	1.7	1.7	1.8	1.9	1.9	2	2	2.1	2.1	2.2	2.2
19	1.1	1.1	1.2	1.3	1.3	1.4	1.5	1.5	1.6	1.6	1.7	1.8	1.8	1.9	1.9	2	2	2	2.1	2.1
20	1.1	1.1	1.2	1.2	1.3	1.4	1.4	1.5	1.5	1.6	1.6	1.7	1.7	1.8	1.8	1.9	1.9	2	2	2
30	1	1.1	1.1	1.1	1.2	1.2	1.2	1.3	1.3	1.3	1.3	1.4	1.4	1.4	1.4	1.5	1.5	1.5	1.5	1.5
40	1	1	1.1	1.1	1.1	1.1	1.1	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.3	1.3
50	1	1	1	1	1	1	1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
60	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
70	1	1	1	1	1	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
80	1	1	1	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.8	0.8	0.8	0.8
90	1	1	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
100	1	1	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.7	0.7

PROCEDURE 34 - TEMPERATURE CALCULATIONS									
Facility:	Cadillac WWTP					Effluent Flow Rate:	4.95 cfs		
Outfall:	001					pH:	7		
Date:	8/1/2023					Drinking Water:	n		
Rule 323.1075(1)(b) - Coldwater streams									
	J	F	M	A	M	J	J	A	S
95% Exc.									
WQS:	38	38	43	54	65	68	68	68	63
TR:	36	36	41	52	63	66	66	66	61
Te:	38	38	43	54	65	68	68	68	63
Rule 323.1075(3)(a) - Warmwater streams, north									
	J	F	M	A	M	J	J	A	S
95% Exc.									
WQS:	38	38	41	56	70	80	83	81	74
TR:	33	33	36	51	65	75	78	76	69
Te:	38	38	41	56	70	80	83	81	74
Rule 323.1075(3)(b) - Warmwater streams, south									
	J	F	M	A	M	J	J	A	S
95% Exc.									
WQS:	41	40	50	63	76	84	85	85	79
TR:	36	35	45	58	71	79	80	80	74
Te:	41	40	50	63	76	84	85	85	79
Rule 323.1075(3)(c) - Warmwater streams, St. Joseph River									
	J	F	M	A	M	J	J	A	S
95% Exc.									
WQS:	50	50	55	65	75	85	85	85	85
TR:	45	45	50	60	70	80	80	80	80
Te:	50	50	55	65	75	85	85	85	85
Rule 323.1070(2)(i) - GLs and Connecting Channels, Detroit River									
	J	F	M	A	M	J	J	A	S
95% Exc.									
WQS:	40	40	45	60	70	75	80	83	80
TR:	37	37	42	57	67	72	77	80	77
Te:	40	40	45	60	70	75	80	83	80

INSTRUCTIONS	(February 2011)
--------------	-----------------

- 1) Information for the top box will be imported from the FACINFO sheet for you.
 - 2) Enter 95% exceedance flows by month in appropriate cells (yellow) according to stream type and location.
 - 3) Delete unneeded rows.
 - 4) Print. ****You will need to select the print area.**** Then, choose File Print Selection OK.
-

[PROCEDURE 34 - FULL TEXT \[click here\]](#)

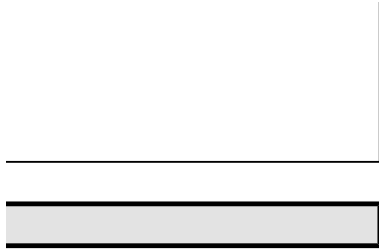
O	N	D
56	48	40
54	46	38
56	48	40

O	N	D
64	49	39
59	44	34
64	49	39

O	N	D
68	55	43
63	50	38
68	55	43

O	N	D
70	60	50
65	55	45
70	60	50

O	N	D
70	55	45
67	52	42
70	55	45



Warm water streams/rivers

Assumptions:

Mixing zone defined by 95% flow, 4 seasons, and effluent flow

water density constant over range of mix temperatures expected

heat capacity of water, C_p , constant over range of mix temperatures expected

no phase changes

$T(WQS)-T_b$; (ΔT)

enter 0.1 to 5.0 F

Q_e (mgd)

requested authorization

Q_b (cfs)

lowest 95% flow, 4 seasons max

Q MBTU/hr

(monthly ave)

Q MBTU/d

(monthly ave)

Cold water streams/rivers

Assumptions:

Mixing zone defined by 95% flow, 4 seasons, and effluent flow

water density constant over range of mix temperatures expected

heat capacity of water, C_p , constant over range of mix temperatures expected

no phase changes

$T(WQS)-T_b$; (ΔT)

enter 0.1 to 2.0 F

Q_e (mgd)

requested authorization

Q_b (cfs)

lowest 95% flow, 4 seasons max

Q MBTU/hr

(monthly ave)

Q MBTU/d

(monthly ave)

Great Lakes Discharges

Assumptions:

Mixing zone defined by 10 parts Lake water, 1 part effluent

water density constant over range of mix temperatures expected

heat capacity of water, Cp, constant over range of mix temperatures expected

no phase changes

T(WQS)-Tb						
	0.5	1	1.5	2	2.5	>=3
Qe (MGD)	Mbtu/hr	Mbtu/hr	Mbtu/hr	Mbtu/hr	Mbtu/hr	Mbtu/hr
0.5	1	2	3	4	5	6
1	2	4	6	8	10	11
1.5	3	6	9	11	14	17
2	4	8	11	15	19	23
2.5	5	10	14	19	24	29
3	6	11	17	23	29	34
3.5	7	13	20	27	33	40
4	8	15	23	31	38	46
4.5	9	17	26	34	43	52
5	10	19	29	38	48	57
10	19	38	57	76	96	115
20	38	76	115	153	191	229
30	57	115	172	229	287	344
40	76	153	229	306	382	459
50	96	191	287	382	478	573
60	115	229	344	459	573	688
70	134	268	401	535	669	803
80	153	306	459	612	765	917
90	172	344	516	688	860	1032
100	191	382	573	765	956	1147
200	382	765	1147	1529	1911	2294
300	573	1147	1720	2294	2867	3440
400	765	1529	2294	3058	3823	4587
500	956	1911	2867	3823	4778	5734
600	1147	2294	3440	4587	5734	6881
700	1338	2676	4014	5352	6689	8027
800	1529	3058	4587	6116	7645	9174
900	1720	3440	5160	6881	8601	10321
1000	1911	3823	5734	7645	9556	11468
1500	2867	5734	8601	11468	14334	17201
2000	3823	7645	11468	15290	19113	22935
2500	4778	9556	14334	19113	23891	28669

Monthly Ave limit would be Mbtu/hr * (Q monthly actual/Q authorized)

DIOXINS and FURANS TEQ and TEC Calculations						
Facility:	Cadillac WWTP	$(TEC)_{TCDD} = \sum [(C)_x (TEF)_x (BEF)_x]$				
Permit #:	MI0020257	$(TEC)_{TCDD} = 2,3,7,8\text{-TCDD toxicity equivalent sum}$				
Date:	8/1/2023	$(C)_x = \text{concentration of congener x in the sample}$				
Reviewer:	Glen Schmitt, Permits Section, WRD,	$(TEF)_x = \text{toxicity equivalency factor for congener x}$				
		$(BEF)_x = \text{bioaccumulation equivalency factor for congener x}$				
Sample or Outfall ID	Congener	Reported Conc	Lab Codes	TEF	TEQ	BEF
	Dioxins (pg/L)					
	2,3,7,8-TCDD			1.0	0	1.0
	1,2,3,7,8-PeCDD			0.5	0	0.9
	1,2,3,4,7,8-HxCDD			0.1	0	0.3
	1,2,3,6,7,8-HxCDD			0.1	0	0.1
	1,2,3,7,8,9-HxCDD			0.1	0	0.1
	1,2,3,4,6,7,8-HpCDD			0.01	0	0.05
	OCDD			0.001	0	0.01
	Total Dioxin Congeners	0		$(TEQ)_{TCDD} =$	0	$(TEC)_{TCDD} =$
	Furans (pg/L)					
	2,3,7,8-TCDF			0.1	0	0.8
	1,2,3,7,8-PeCDF			0.05	0	0.2
	2,3,4,7,8-PeCDF			0.5	0	1.6
	1,2,3,4,7,8-HxCDF			0.1	0	0.08
	1,2,3,6,7,8-HxCDF			0.1	0	0.2
	2,3,4,6,7,8-HxCDF			0.1	0	0.7
	1,2,3,7,8,9-HxCDF			0.1	0	0.6
	1,2,3,4,6,7,8-HpCDF			0.01	0	0.01
	1,2,3,4,7,8,9-HpCDF			0.01	0	0.4
	OCDF			0.001	0	0.02
	Total Furan Congeners	0		$(TEQ)_{TCDF} =$	0	$(TEC)_{TCDF} =$
						Total
Sample or Outfall ID	Congener	Reported Conc	Lab Codes	TEF	TEQ	BEF
	Dioxins (pg/L)					
	2,3,7,8-TCDD			1.0	0	1.0
	1,2,3,7,8-PeCDD			0.5	0	0.9
	1,2,3,4,7,8-HxCDD			0.1	0	0.3
	1,2,3,6,7,8-HxCDD			0.1	0	0.1
	1,2,3,7,8,9-HxCDD			0.1	0	0.1
	1,2,3,4,6,7,8-HpCDD			0.01	0	0.05
	OCDD			0.001	0	0.01

Total Dioxin Congeners		0	(TEQ)_{TCDD} =	0	(TEC)_{TCDD} =
Furans (pg/L)					
2,3,7,8-TCDF		0.1	0	0.8	
1,2,3,7,8-PeCDF		0.05	0	0.2	
2,3,4,7,8-PeCDF		0.5	0	1.6	
1,2,3,4,7,8-HxCDF		0.1	0	0.08	
1,2,3,6,7,8-HxCDF		0.1	0	0.2	
2,3,4,6,7,8-HxCDF		0.1	0	0.7	
1,2,3,7,8,9-HxCDF		0.1	0	0.6	
1,2,3,4,6,7,8-HpCDF		0.01	0	0.01	
1,2,3,4,7,8,9-HpCDF		0.01	0	0.4	
OCDF		0.001	0	0.02	
Total Furan Congeners		0	(TEQ)_{TCDF} =	0	(TEC)_{TCDF} =
					Tot
		MI WATER QUALITY STDS.	HNV Drink = Non-Drink	HCV Drink = Non-Drink	
		2,3,7,8-TCDD	---	---	
		Total TECs	6.7 x 10 ⁻⁸ ug/l	8.6 x 10 ⁻⁹ ug/l	

DIOXINS and FURANS TEQ and TEC Calcul						
Facility:	Cadillac WWTP	$(TEC)_{TCDD} = \sum [(C)_x (TEF)_x (BEF)_x]$				
Permit #:	MI0020257	$(TEC)_{TCDD} = 2,3,7,8\text{-TCDD toxicity equiv}$				
Date:	8/1/2023	$(C)_x = \text{concentration of congener x in the}$				
Reviewer:	Glen Schmitt, Permits Section, WRD,	$(TEF)_x = \text{toxicity equivalency factor for c}$				
		$(BEF)_x = \text{bioaccumulation equivalency fa}$				
Sample or Outfall ID	Congener	< Reported Conc	Lab Codes	TEF	TEQ	BEF
Dioxins (pg/L)						
	2,3,7,8-TCDD			1.0	0	1.0
	1,2,3,7,8-PeCDD			0.5	0	0.9
	1,2,3,4,7,8-HxCDD			0.1	0	0.3
	1,2,3,6,7,8-HxCDD			0.1	0	0.1
	1,2,3,7,8,9-HxCDD			0.1	0	0.1
	1,2,3,4,6,7,8-HpCDD			0.01	0	0.05
	OCDD			0.001	0	0.01
Total Dioxin Congeners		0	(TEQ)_{TCDD} =	0	(TEC)_{TCDD} =	
Furans (pg/L)						
	2,3,7,8-TCDF			0.1	0	0.8
	1,2,3,7,8-PeCDF			0.05	0	0.2

	2,3,4,7,8-PeCDF		0.5	0	1.6
	1,2,3,4,7,8-HxCDF		0.1	0	0.08
	1,2,3,6,7,8-HxCDF		0.1	0	0.2
	2,3,4,6,7,8-HxCDF		0.1	0	0.7
	1,2,3,7,8,9-HxCDF		0.1	0	0.6
	1,2,3,4,6,7,8-HpCDF		0.01	0	0.01
	1,2,3,4,7,8,9-HpCDF		0.01	0	0.4
	OCDF		0.001	0	0.02
	Total Furan Congeners	0	(TEQ) _{TCDF} =	0	(TEC) _{TCDF} =
					Tot
Sample or			Lab		
	Congener	< Reported	Codes	TEF	TEQ
Outfall ID		Conc			BEF
	Dioxins (pg/L)				
	2,3,7,8-TCDD			1.0	0
	1,2,3,7,8-PeCDD			0.5	0
	1,2,3,4,7,8-HxCDD			0.1	0
	1,2,3,6,7,8-HxCDD			0.1	0
	1,2,3,7,8,9-HxCDD			0.1	0
	1,2,3,4,6,7,8-HpCDD			0.01	0
	OCDD			0.001	0
	Total Dioxin Congeners	0	(TEQ) _{TCDD} =	0	(TEC) _{TCDD} =
	Furans (pg/L)				
	2,3,7,8-TCDF			0.1	0
	1,2,3,7,8-PeCDF			0.05	0
	2,3,4,7,8-PeCDF			0.5	0
	1,2,3,4,7,8-HxCDF			0.1	0
	1,2,3,6,7,8-HxCDF			0.1	0
	2,3,4,6,7,8-HxCDF			0.1	0
	1,2,3,7,8,9-HxCDF			0.1	0
	1,2,3,4,6,7,8-HpCDF			0.01	0
	1,2,3,4,7,8,9-HpCDF			0.01	0
	OCDF			0.001	0
	Total Furan Congeners	0	(TEQ) _{TCDF} =	0	(TEC) _{TCDF} =
					Tot
		MI WATER QUALITY STDS.	HNV Drink = Non-Drink	HCV Drink = Non-Drink	
		2,3,7,8-TCDD	---	---	
		Total TECs	6.7 x 10 ⁻⁸ ug/l	8.6 x 10 ⁻⁹ ug/l	

DIOXINs and FURANs TEQ and TEC Calcul

Facility:	Cadillac WWTP	$(TEC)_{TCDD} = \sum [(C)_x (TEF)_x (BEF)_x]$
Permit #:	MI0020257	$(TEC)_{TCDD} = 2,3,7,8\text{-TCDD toxicity equiv}$
Date:	8/1/2023	$(C)_x = \text{concentration of congener x in the}$
Reviewer:	Glen Schmitt, Permits Section, WRD,	$(TEF)_x = \text{toxicity equivalency factor for c}$
		$(BEF)_x = \text{bioaccumulation equivalency fa}$

Sample or Outfall ID	Congener	< Reported Conc	Lab Codes	TEF	TEQ	BEF
	Dioxins (pg/L)					
	2,3,7,8-TCDD			1.0	0	1.0
	1,2,3,7,8-PeCDD			0.5	0	0.9
	1,2,3,4,7,8-HxCDD			0.1	0	0.3
	1,2,3,6,7,8-HxCDD			0.1	0	0.1
	1,2,3,7,8,9-HxCDD			0.1	0	0.1
	1,2,3,4,6,7,8-HpCDD			0.01	0	0.05
	OCDD			0.001	0	0.01
	Total Dioxin Congeners	0		$(TEQ)_{TCDD} =$	0	$(TEC)_{TCDD} =$
	Furans (pg/L)					
	2,3,7,8-TCDF			0.1	0	0.8
	1,2,3,7,8-PeCDF			0.05	0	0.2
	2,3,4,7,8-PeCDF			0.5	0	1.6
	1,2,3,4,7,8-HxCDF			0.1	0	0.08
	1,2,3,6,7,8-HxCDF			0.1	0	0.2
	2,3,4,6,7,8-HxCDF			0.1	0	0.7
	1,2,3,7,8,9-HxCDF			0.1	0	0.6
	1,2,3,4,6,7,8-HpCDF			0.01	0	0.01
	1,2,3,4,7,8,9-HpCDF			0.01	0	0.4
	OCDF			0.001	0	0.02
	Total Furan Congeners	0		$(TEQ)_{TCDF} =$	0	$(TEC)_{TCDF} =$
						Total

Sample or Outfall ID	Congener	< Reported Conc	Lab Codes	TEF	TEQ	BEF
	Dioxins (pg/L)					
	2,3,7,8-TCDD			1.0	0	1.0
	1,2,3,7,8-PeCDD			0.5	0	0.9
	1,2,3,4,7,8-HxCDD			0.1	0	0.3
	1,2,3,6,7,8-HxCDD			0.1	0	0.1
	1,2,3,7,8,9-HxCDD			0.1	0	0.1
	1,2,3,4,6,7,8-HpCDD			0.01	0	0.05
	OCDD			0.001	0	0.01
	Total Dioxin Congeners	0		$(TEQ)_{TCDD} =$	0	$(TEC)_{TCDD} =$

Furans (pg/L)				
2,3,7,8-TCDF	0.1	0	0.8	
1,2,3,7,8-PeCDF	0.05	0	0.2	
2,3,4,7,8-PeCDF	0.5	0	1.6	
1,2,3,4,7,8-HxCDF	0.1	0	0.08	
1,2,3,6,7,8-HxCDF	0.1	0	0.2	
2,3,4,6,7,8-HxCDF	0.1	0	0.7	
1,2,3,7,8,9-HxCDF	0.1	0	0.6	
1,2,3,4,6,7,8-HpCDF	0.01	0	0.01	
1,2,3,4,7,8,9-HpCDF	0.01	0	0.4	
OCDF	0.001	0	0.02	
Total Furan Congeners	0	(TEQ)_{TCDF} =	0	(TEC)_{TCDF} =
				Tot
		MI WATER QUALITY STDS.	HNV Drink = Non-Drink	HCV Drink = Non-Drink
		2,3,7,8-TCDD	---	---
		Total TECs	6.7 x 10 ⁻⁸ ug/l	8.6 x 10 ⁻⁹ ug/l

DIOXINS and FURANS TEQ and TEC Calculations						
Facility:	Cadillac WWTP	(TEC) _{TCDD} = sum[(C) _x (TEF) _x (BEF) _x]				
Permit #:	MI0020257	(TEC) _{TCDD} = 2,3,7,8-TCDD toxicity equiv				
Date:	8/1/2023	(C) _x = concentration of congener x in the				
Reviewer:	Glen Schmitt, Permits Section, WRD,	(TEF) _x = toxicity equivalency factor for c				
		(BEF) _x = bioaccumulation equivalency fa				
Sample or Outfall ID	Congener	Reported Conc	Lab Codes	TEF	TEQ	BEF
Dioxins (pg/L)						
	2,3,7,8-TCDD			1.0	0	1.0
	1,2,3,7,8-PeCDD			0.5	0	0.9
	1,2,3,4,7,8-HxCDD			0.1	0	0.3
	1,2,3,6,7,8-HxCDD			0.1	0	0.1
	1,2,3,7,8,9-HxCDD			0.1	0	0.1
	1,2,3,4,6,7,8-HpCDD			0.01	0	0.05
	OCDD			0.001	0	0.01
	Total Dioxin Congeners	0		(TEQ)_{TCDD} =	0	(TEC)_{TCDD} =
Furans (pg/L)						
	2,3,7,8-TCDF			0.1	0	0.8
	1,2,3,7,8-PeCDF			0.05	0	0.2
	2,3,4,7,8-PeCDF			0.5	0	1.6
	1,2,3,4,7,8-HxCDF			0.1	0	0.08

	1,2,3,6,7,8-HxCDF		0.1	0	0.2
	2,3,4,6,7,8-HxCDF		0.1	0	0.7
	1,2,3,7,8,9-HxCDF		0.1	0	0.6
	1,2,3,4,6,7,8-HpCDF		0.01	0	0.01
	1,2,3,4,7,8,9-HpCDF		0.01	0	0.4
	OCDF		0.001	0	0.02
	Total Furan Congeners	0	(TEQ) _{TCDF} =	0	(TEC) _{TCDF} =
					Tot
Sample or		Reported	Lab		
	Congener	< Conc		TEF	TEQ
Outfall ID			Codes		BEF
	Dioxins (pg/L)				
	2,3,7,8-TCDD			1.0	0
	1,2,3,7,8-PeCDD			0.5	0
	1,2,3,4,7,8-HxCDD			0.1	0
	1,2,3,6,7,8-HxCDD			0.1	0
	1,2,3,7,8,9-HxCDD			0.1	0
	1,2,3,4,6,7,8-HpCDD			0.01	0
	OCDD			0.001	0
	Total Dioxin Congeners	0	(TEQ) _{TCDD} =	0	(TEC) _{TCDD} =
	Furans (pg/L)				
	2,3,7,8-TCDF			0.1	0
	1,2,3,7,8-PeCDF			0.05	0
	2,3,4,7,8-PeCDF			0.5	0
	1,2,3,4,7,8-HxCDF			0.1	0
	1,2,3,6,7,8-HxCDF			0.1	0
	2,3,4,6,7,8-HxCDF			0.1	0
	1,2,3,7,8,9-HxCDF			0.1	0
	1,2,3,4,6,7,8-HpCDF			0.01	0
	1,2,3,4,7,8,9-HpCDF			0.01	0
	OCDF			0.001	0
	Total Furan Congeners	0	(TEQ) _{TCDF} =	0	(TEC) _{TCDF} =
					Tot
		MI WATER QUALITY STDS.	HNV Drink = Non-Drink	HCV Drink = Non-Drink	
		2,3,7,8-TCDD	---	---	
		Total TECs	6.7 x 10 ⁻⁸ ug/l	8.6 x 10 ⁻⁹ ug/l	

Factor		
Valence concentration Discharge Congener x Factor for congener x		
TEC	Total TEC	Notes
	PCDD TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0	PCDF TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0		
0		
0		
0		
0	0	
Total TECs = --- > MWQS		
TEC	Total TEC	Notes
	PCDD TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		

0		
0	PCDF TECs	
0	(%)	
0	---	
0		
0		
0		
0		
0		
0		
0		
0		Cumulative TECs
0	0	0
Total TECs = --- > MWQS		
WV		Enforceable
-----		WQ Value (pg/l)
3.1 x 10 ⁻⁹ ug/l		0.0031
---		0.0086

ator		
alence concentration discharge ongener x ctor for congener x		
TEC	Total TEC	Notes
0	PCDD TECs	
0	(%)	
0	---	
0		
0		
0		
0		
0		
0		
0	PCDF TECs	
0	(%)	

valence concentration
 discharge
 ongener x
 ctor for congener x

TEC	Total TEC	Notes
	PCDD TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0		
	PCDF TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0		
0		
0		
0		
0	0	
Total TECs = --- > MWQS		
TEC	Total TEC	Notes
	PCDD TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0		

0		
0		
0		
0		
0		
0		
0		
0	0	
Total TECs = --- > MWQS		
TEC	Total TEC	Notes
0	PCDD TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0	PCDF TECs (%) ---	
0		
0		
0		
0		
0		
0		
0		
0		
0		
0		
0		Cumulative TECs
0	0	0
Total TECs = --- > MWQS		
WV -----	Enforceable WQ Value (pg/l)	
3.1 x 10 ⁻⁹ ug/l	0.0031	
---	0.0086	

Nutrient Checklist

Receiving water name: Clam River

Is the receiving water a stream: ☐ lake: ☐ impoundment: ☐

If a stream, is there a lake or impoundment within 20 miles downstream of the discharge?

Yes: ☐ No: ☐

If yes, describe:

Lake/impoundment characteristics (add rows to address multiple lakes/impoundments):

Name:

Size (acres): ☐ Avg Depth (ft): ☐ Retention (days): ☐

Nutrient concerns, data, current trophic status, etc.:

For the receiving water or a waterbody of concern within 20 miles downstream:

Has a TMDL been established for nutrient concerns? Yes: ☐ No: ☐

Explain TMDL restrictions relating to this facility or downstream waterbodies, if any:

Is the waterbody included on the nonattainment list for nutrient concerns? Yes: ☐

No: ☐

Are there any NPS-related projects (e.g. 319, CMI) to reduce nutrient inputs in the vicinity or downstream of the discharge?

Yes: ☐ No: ☐

If yes, describe:

Are there other point source discharges of nutrients that need to be considered in this review?

If so, identify:

BACKGROUND

Concentrati mg/l
Flow cfs

DISCHARGE

Concentrati mg/l
Flow cfs

Conc mix mg/l

Solve for k based on 'half-life'

HALF-LIFE

C(t) 1 mg/l

Co 2 mg/l

k /d

Days to Y mg/l from initial X mg/l

C(t) 2 mg/l

Co 40 mg/l

k #DIV/0! /d

t d

MASS BALANCE**BACKGROUND CONDITIONS**

Flow (cfs) TP (mg/l)

REACHES AND INFLOWS

Discharge	Inflow Q (cfs)	Inflow TP (mg/l)	Total Q (cfs)
1			0.00
2			0.00
3			0.00
4			0.00
5			0.00
6			0.00
7			0.00
8			0.00
9			0.00
10			0.00



[illegible]

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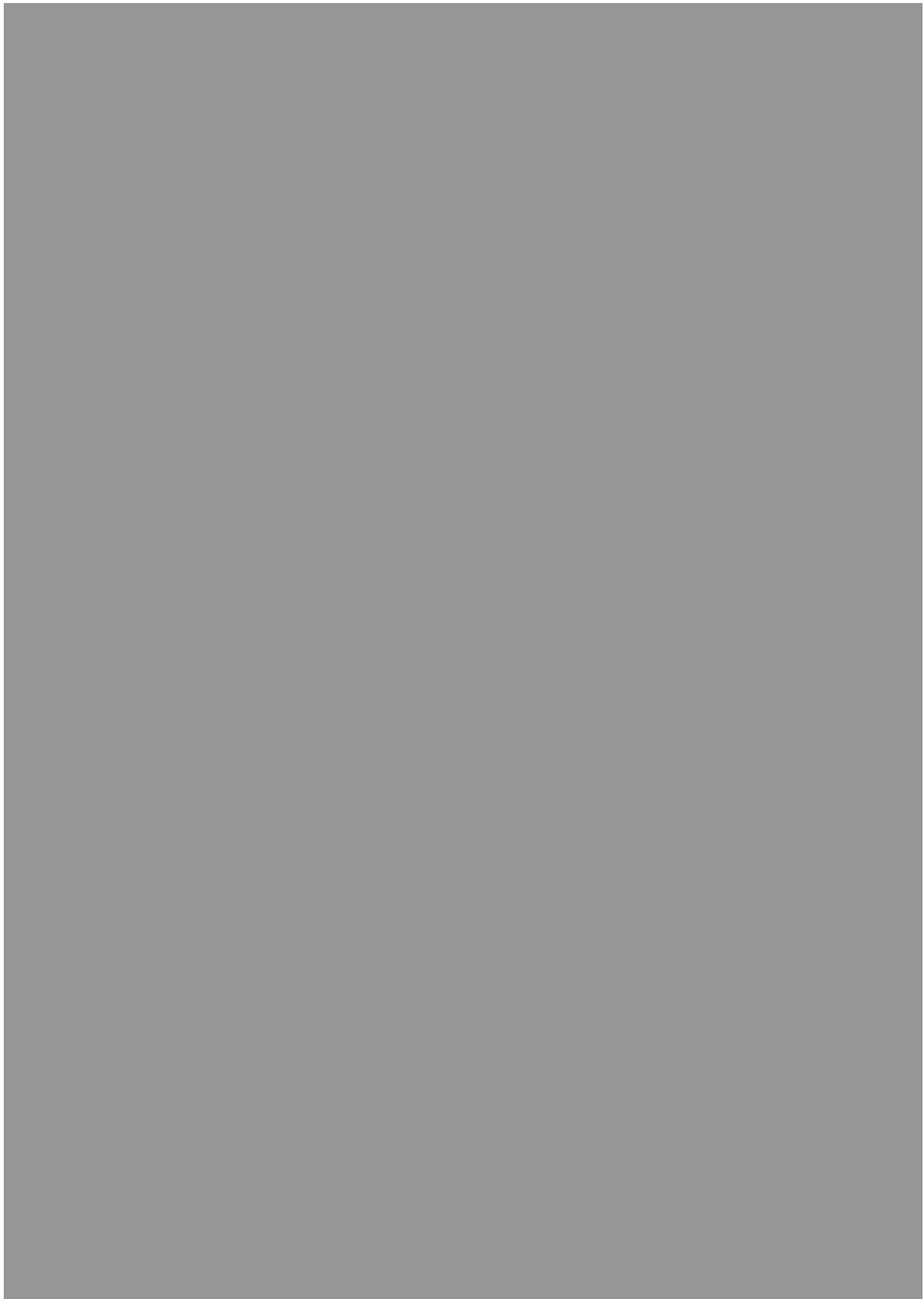
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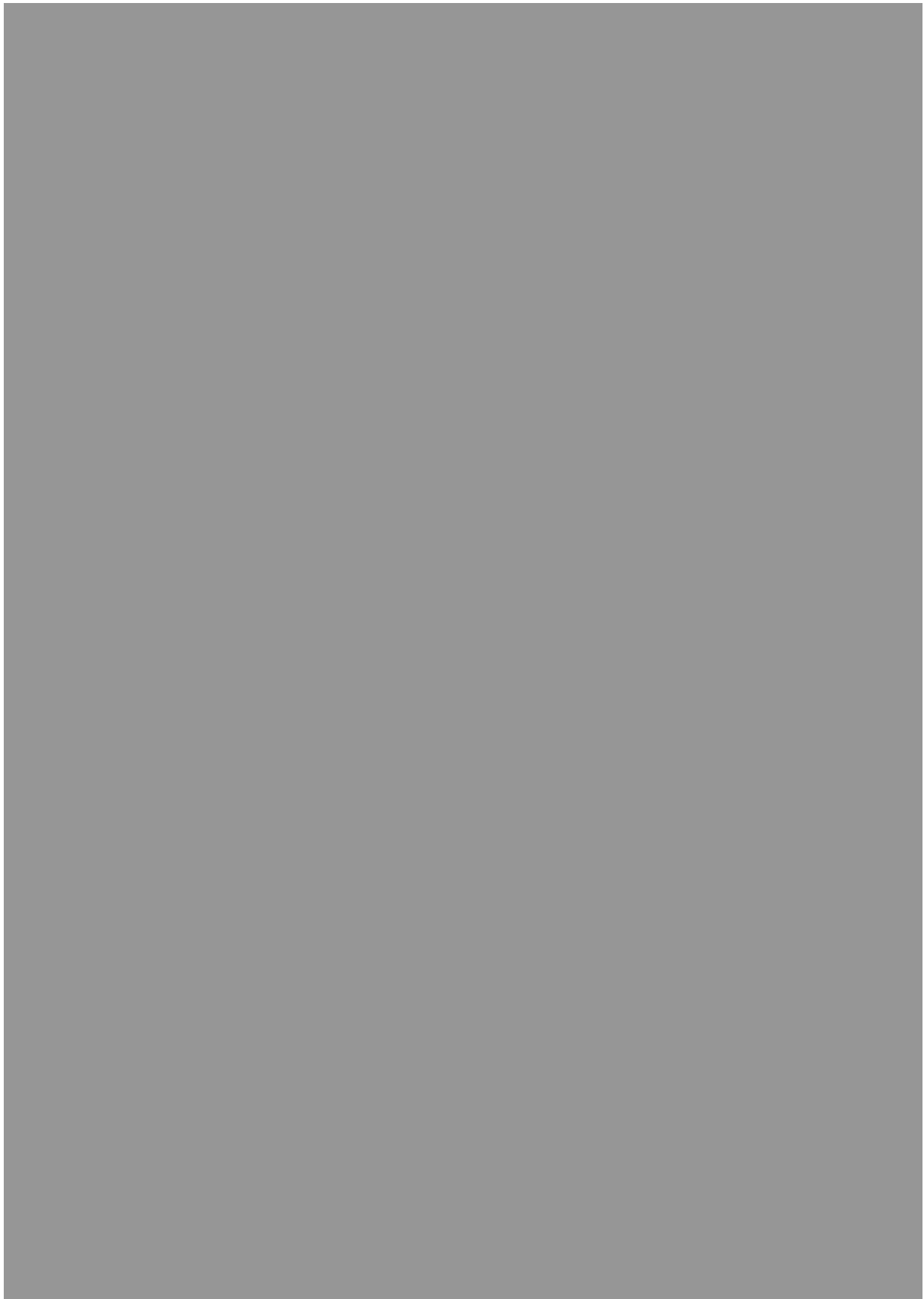
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1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055	2056	2057	2058	2059	2060	2061	2062	2063	2064	2065	2066	2067	2068	2069	2070	2071	2072	2073	2074	2075	2076	2077	2078	2079	2080	2081	2082	2083	2084	2085	2086	2087	2088	2089	2090	2091	2092	2093	2094	2095	2096	2097	2098	2099	2100	

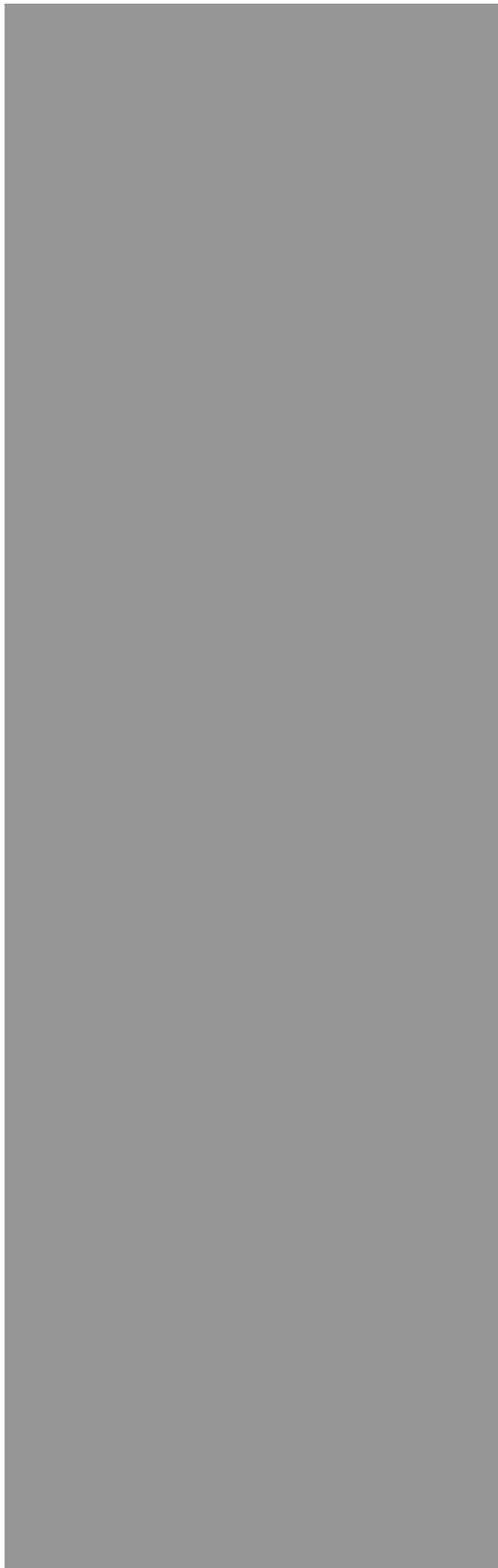


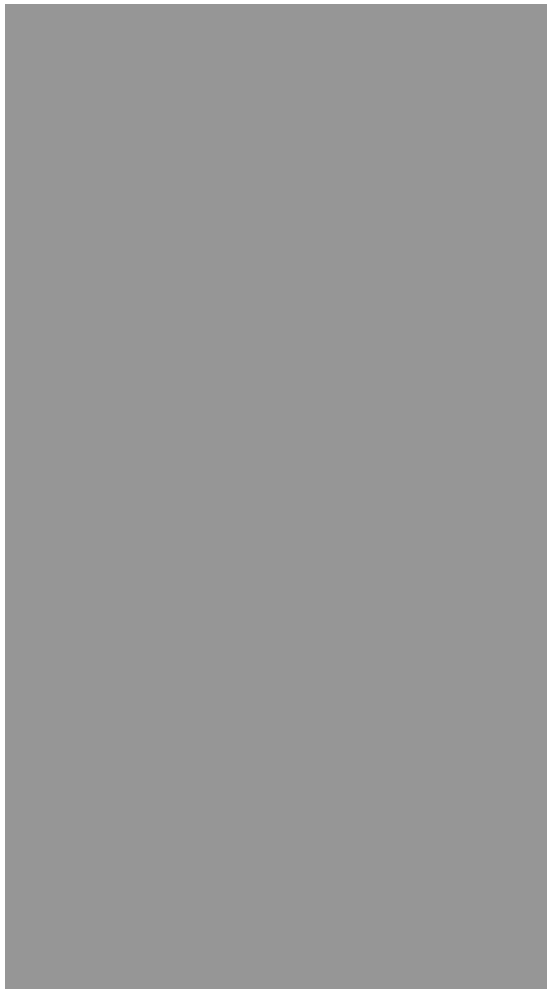












Un-ionized Hydrogen Sulfide Concentration Calculations	
Total Sulfide Concentration (mg/L)	
pH	
Conductivity	
Temperature (Celsius)	
Ionic Strength	0
A - Debye-Huckel Constant	0.4891618206
γ_{H^+} - Monovalent Ion Activity Coefficient	0
pK1 - Dissociation Constant	7.3645211979
K'1 - Conditional Ionization Constant	4.319950814701E-008
[H ⁺] - Hydrogen Ion Concentration	1
[H ₂ S] (M)	0
[H ₂ S] (mg/L)	0

Un-ionized Hydrogen Sulfide Concentration Calculations	
H2S Acute WQBEL ug/L	6.4
pH	
Conductivity	
Temperature (Celsius)	
Ionic Strength	0
A - Debye-Huckel Constant	0.4891618206
γ_{H^+} - Monovalent Ion Activity Coefficient	0
pK1 - Dissociation Constant	7.3645211979
K'1 - Conditional Ionization Constant	4.319950814701E-008
[H ⁺] - Hydrogen Ion Concentration	1
Total Sulfide Acute WQBEL (ug/L)	6.40
Total Sulfide Acute WQBEL (lbs/day)	0.1708032074

Use worst case conditions to calculate dissolved sulfide WQBEL
pH = 6.5
conductivity = minimum effluent concentration if effluent dominated
or minimum receiving waterbody conductivity
Temp = use lowest WQS or lowest effluent temp if effluent dominated

Un-ionized Hydrogen Sulfide Concentration Calculations	
H2S Chronic WQBEL ug/L	
pH	
Conductivity	
Temperature (Celsius)	
Ionic Strength	0
A - Debye-Huckel Constant	0.4891618206
γ_{H^+} - Monovalent Ion Activity Coefficient	0
pK1 - Dissociation Constant	7.3645211979
K'1 - Conditional Ionization Constant	4.319950814701E-008
[H ⁺] - Hydrogen Ion Concentration	1
Total Sulfide Chronic WQBEL (ug/L)	0
Total Sulfide Chronic WQBEL (lbs/day)	0

PUBLIC WATER SUPPLY INTAKE**GREAT LAKES AND CONNECTING**

WATER SUPPLY	COUNTY	LATITUDE
--------------	--------	----------

DETROIT RIVER

Detroit Water and Sewerage Dept.	Wayne	42.35018
Detroit Water and Sewerage Dept.	Wayne	42.22732
Detroit Water and Sewerage Dept.	Wayne	42.35000

LAKE ERIE

City of Monroe / Frenchtown Township	Monroe	41.93667
City of Monroe / Frenchtown Township	Monroe	41.94173

LAKE HURON

Alpena	Alpena	45.04641
Alpena	Alpena	45.04613
Bay City	Bay	43.71722
Bay City	Bay	43.67333
Captain Morgan's Bar and Grill	Huron	44.0561
Caseville	Huron	43.94861
DeTour Village	Chippewa	45.9997
Detroit Water and Sewerage Dept.	St. Clair	43.12611
Harbor Beach	Huron	43.86167
Harbor Beach	Huron	43.85833
Huron Regional Water Authority	Huron	44.04933
Huron Regional Water Authority	Huron	44.054667
Huron Shore Regional Utility Authority	Iosco	44.26944
Lexington	Sanilac	43.26444
Mackinac Island	Mackinac	45.85417
Saginaw-Midland	Arenac	44.10831
Saginaw-Midland	Arenac	44.10782
St. Ignace	Mackinac	45.8586
Whalen's Grindstone Shores	Huron	44.05611

LAKE MICHIGAN

Benton Harbor	Berrien	42.13055
Bridgman	Berrien	41.94306
Charlevoix	Charlevoix	45.32222
Escanaba	Delta	45.74247
Fayette State Park	Delta	45.71782
Fayette State Park	Delta	45.71822
Gladstone	Delta	45.84357

Grand Rapids	Kent	42.9625
Grand Rapids\	Kent	42.97083
Holland	Ottawa	42.8
Lake Township	Berrien	41.96619
Ludington	Mason	43.96187
Ludington	Mason	43.96024
Menominee	Menominee	45.13778
Muskegon	Muskegon	43.2075
Muskegon Heights	Muskegon	43.180591
Muskegon Heights	Muskegon	43.181891
New Buffalo	Berrien	41.80194
NW Ottawa Co Water System - Grand Haven	Ottawa	43.049688
NW Ottawa Co Water System - Grand Haven	Ottawa	43.051137
South Haven	Van Buren	42.39886
St. Joseph	Berrien	42.10115
Traverse City	Grand Traverse	44.7694
Wyoming	Kent	42.85425

LAKE ST. CLAIR

Grosse Pointe Farms/Highland Park	Wayne	42.40313
Ira Township	St. Clair	42.68056
Mount Clemens	Macomb	42.562567
Mount Clemens	Macomb	42.55825
New Baltimore	Macomb	42.67778

ST. CLAIR RIVER

Algonac	St. Clair	42.62083
Marine City	St. Clair	42.71653
Marysville	St. Clair	42.90694
Port Huron	St. Clair	42.986666
St. Clair	St. Clair	42.82533
St. Clair Water & Sewer Authority	St. Clair	42.75554
The Old Club - Harsen's Island	St. Clair	42.53832

LAKE SUPERIOR

Baraga	Baraga	46.77073
Isle Royale Mott Island Intake	Keweenaw	48.14498
Isle Royale Rock Harbor Intake	Keweenaw	48.10499
Isle Royale Windigo Intake	Keweenaw	47.90944
Keweenaw Bay Intake	Baraga	46.80639
L'Anse	Baraga	46.78648
Marquette	Marquette	46.54463
Ontonagon	Ontonagon	46.83585
St Mary's River Intake	Chippewa	46.49139

INLAND LAKES

Water Supply	County	Source
Adrian	Lenawee	Lake Adrian

RIVERS

Water Supply	County	Source
Alma	Gratiot	Pine River
Ann Arbor	Washtenaw	Huron River
Blissfield	Lenawee	River Raisin
Deerfield	Lenawee	River Raisin
Flint (Emergency Use Only)	Genesee	Flint River
Manistique	Schoolcraft	Indian River

DAT:md 8/09/89 , 9/19/89, 9/06/95 revised

BBB: 8/24/98 revised

MDW: 8/31/09 Updated list per Kris Phillip (SDWIS database)

S IN MICHIGAN

NG WATERS

LONGITUDE	ADDITIONAL INFO
-82.95545	Belle Isle
-83.12873	Southwest Water Plant
-82.95556	Water Works Park
-83.23862	W.L. Lepage Raw Water Pumping Plant - 30"
-83.24994	W.L. Lepage Raw Water Pumping Plant - 42"
-83.43668	Thunder Bay Intake 1
-83.44119	Thunder Bay Intake 2
-83.9014	Saginaw Bay Intake - New
-83.8975	Saginaw Bay Intake - Old
-82.89579	
-83.28195	
-83.8983	DeTour Passage Intake
-82.39084	Lake Huron WTP Intake
-82.64667	Emergency Intake - 20"
-82.64585	Harbor Beach Water Works - 30"
-82.993167	Port Austin - Emergency Intake
-83.0025	Port Austin
-83.41472	formerly Tawas Utilities Authority
-82.51306	
-84.59999	
-83.54999	Lake Huron Intake #2
-83.52638	Lake Huron Intake #1
-84.7039	Straits of Mackinac Intake
-82.89569	
-86.48612	
-86.5847	
-85.27028	
-87.03574	
-86.66443	
-86.66382	
-86.99874	

-86.23751	South Intake
-86.24584	North Intake
-86.22785	
-86.58248	
-86.46622	New Intake
-86.46921	Old Intake
-87.59223	
-86.34751	
-86.326	
-86.327208	
-86.76444	
-86.248658	Intake 4 - South Buried Intake
-86.250607	Intake 3 - North Buried Intake
-86.30506	
-86.49893	
-85.53791	
-86.22696	
-82.87852	
-82.67083	
-82.8366	Emergency Intake
-82.82481	
-82.73195	
-82.5264	
-82.49034	
-82.4639	
-82.424722	
-82.48132	
-82.47361	East China Township
-82.66715	
-88.48969	
-88.48168	
-88.54204	
-89.16028	
-88.42362	
-88.44263	
-87.36575	
-89.574333	
-84.4214	

Placeholder for future Silver Translator guidance....

The PEL sheet requires a translator for Silver in the PEL 1 worksheet.
The translator is based on the type of treatment available.

As a result, use the following translators based on the facility type:

WWTPs use a translator of 20

Other Facilities (Industrial, Power Plants, Paper Plants, etc) use a translator of 10

Venting Groundwater Reviews use a translator of 5